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An Analysis of Independent Power Projects in Africa: understanding development and investment outcomes

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Acronyms

ABB	Asea Brown Boveri
ADB	Asian Development Bank
AFD	Agence Francaise de Developpement
AfDB	African Development Bank
AFUDC	Allowance for funds used during construction
AKFED	Aga Khan Fund for Economic Development
ATJL	Artumas Tanzania (Jersey) Limited
BFI	Bilateral finance institution
BOAD	West African Bank for Development
BOO	Build own operate
BOOT	Build own operate transfer
BOT	Build operate transfer
BTO	Build transfer operate
CCGT	Combined cycle gas turbines
CDC	Commonwealth Development Corporation
CED	Compagnie Eolienne de Detroit
CIPREL	Compagnie Ivoirienne de Production d' Electricité
CMS	Consumer Michigan Services
COD	Commercial operation date
CPI	Consumer Price Index
CUE	Cost of unserved energy
DEG	German Investment & Development Corporation
DFI	Development finance institution
DFID	Department for International Development
DRC	Democratic Republic of Congo
EAP	East Asia and Pacific
ECA	Europe and Central Asia
EDF	Electricité de France
EEHC	Egyptian Electricity Holding Company
EETC	Egyptian Electricity Transmission Company
EIB	European Investment Bank
EPC	Engineering, procurement, and construction
ERA	Electricity Regulatory Authority (of Egypt)
ERB	Electricity Regulatory Board (of Kenya)
ESI	Electricity supply industry
FDI	Foreign direct investment
FMO	Netherlands Development Company
GDP	Gross domestic product
GNI	Gross national income
GoE	Government of Egypt
HFO	Heavy fuel oil
IADB	Inter-American Development Bank
ICB	International competitive bid
IDA	International Development Association
IFC	International Finance Corporation
IMF	International Monetary Fund
IPP	Independent power project
IPS	Industrial Promotion Services
IPTL	Independent Power Tanzania Limited
ISP	Independent service provider
IT	Information technology
JV	Joint venture

JBIC	Japan Bank for International Cooperation
KenGen	Kenya Generating Company Limited
km	kilometer
KPLC	Kenya Power and Light Company
kV	kilovolt
kVa	kilovolt ampere
kW	kilowatt
kWh	kilowatt hour
LAC	Latin America and Caribbean
LNG	Liquefied natural gas
MDG	Millennium development goal
MEP	Mtwara Energy Project
MFI	Mutilateral finance institution
MIR	Management Programme in Infrastructure Reform and Regulation
MMBtu	Million British thermal units
MW	Megawatt
O&M	Operating and maintenance
OCGT	Open cycle gas turbine
OECD	Organization for Economic Cooperation and Development
ONE	Office Nationale de l'Electricité (of Morocco)
OPIC	Overseas Private Investment Corporation (of the United States)
PEL	Pendekar Energy Limited
PESD	Program on Energy and Sustainable Development
PHCN	Power Holding Company of Nigeria
PPA	Power purchase agreement
PPI	Private participation in infrastructure
PRG	Partial risk guarantee
PRI	Political risk insurance
PROPARCO	Promotion et Participation pour la Cooperation economique
PSEG	Public Service Enterprise Group
ROE	Return on equity
RSA	Republic of South Africa
SA	South Asia
SAPP	Southern African Power Pool
SEEB	Societé d'Electricité d'El Biban
Sida	Swedish International Development Cooperation Agency
SSA	Sub-Saharan Africa
STEG	Societé Tunisienne d'Electricité et du Gaz
TANESCO	Tanzania Electric Supply Company Limited
TAQA	Abu Dhabi National Energy Company
Tcf	Trillion cubic feet
TDFL	Tanzania Development Finance Company Limited
TPDC	Tanzania Petroleum Development Corporation
TWh	Terawatt hour
UK	United Kingdom
USA	United States of Amerca
VAT	Value added tax
VIP	VIP Engineering Limited
VRA	Volta River Authority
WDI	World Development Indicators
YFP	Yinka Folawiyo Power Limited

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Abstract

This study analyzes the outcomes of African independent power projects (IPPs). IPPs in Africa arose through the need to attract investment for new electricity generating capacity. They were also seen as a way to introduce the private sector into the electric generation sector and thereby improve technical and financial performance. Approximately two dozen such projects have taken root to date, concentrated in mainly eight countries. Outcomes have been varied with more balanced outcomes perceived in the North African region than across Sub-Saharan Africa. The main contribution of this study is in identifying a series of contributing elements to success, which appear to improve the probability for more balanced investment and development outcomes and ultimately more sustainable IPPs.

1. Introduction

At the beginning of the 1990s, virtually all major power generation throughout Africa was financed by public coffers, including via concessionary loans from development finance institutions (DFI).[1] These publicly financed generation assets were considered one of the core elements in state-owned, vertically integrated power systems. In the early 1990s, however, a confluence of factors brought about a significant change in business. With the main drivers identified as insufficient public funds for new generation and decades of poor performance by state-run utilities, developing countries throughout Africa began to adopt a new 'standard' model for their power systems, influenced by pioneering reformers in the United States of America (USA), the United Kingdom (UK), Chile and Norway.[2] Urged on by multilateral and bilateral development institutions, which largely withdrew from funding state-owned projects, a number of countries adopted plans to unbundle their power systems and introduce private participation and competition. Independent power projects (IPP), namely privately-financed, greenfield generation, supported by non-recourse or limited recourse loans, with long-term power purchase agreements (PPA) with the state utility or another off-taker, became a priority within overall power sector reform (World Bank 1993:45,51; World Bank and USAID 1994:1). IPPs were considered a quick and relatively easy fix to persistent supply constraints, and could also potentially serve to benchmark state-owned supply and gradually introduce competition (APEC Energy Working Group 1997). IPPs could be undertaken before sector unbundling. An independent regulator was also not a prerequisite since the PPA laid down a form of regulation by contract.

In 1994, Cote d'Ivoire became among the first African countries to attract a foreign-led IPP to sell power to the grid under long-term contracts with the state utility. Cote d'Ivoire kicked

[1] The term 'multilateral development institutions' (MDI) is used throughout this paper to refer to such organizations as the World Bank Group (including the International Development Association (IDA), International Finance Corporation (IFC), Multilateral Investment Guarantee Agency (MIGA) and International Bank for Reconstruction and Development (IBRD)), Inter-American Development Bank (IADB), Asian Development Bank (ADB) and other regional banks. The term 'bilateral development institutions' (BDI) is used to refer to bilateral funding agencies, such as the UK Department for International Development (DFID), the Commonwealth Development Corporation (CDC), the Investment and Promotions Company for Economic Cooperation (PROPARCO) and the United States Agency for International Development (USAID), which are distinct from Export Credit Agencies (ECA), for which funding is directly tied to exports. The term 'development finance institutions' (DFI) will be used throughout to refer to all such multilateral and bilateral organizations.

[2] The standard model for power sector reform has been roughly defined as a series of steps that move vertically-integrated utilities toward competition, and generally include the following activities, in the following order: corporatization, commercialization, the passage of the requisite energy legislation, the establishment of an independent regulator, the introduction of IPPs, restructuring/unbundling, divestiture of generation and distribution assets and the introduction of competition (Bacon 1999:4) (Adamantiades, Besant-Jones et al. 1995:6-7; Besant-Jones 2006:11; Williams and Ghanadan 2006:822). Important to note that although this model, which was based largely on the early power sector reforms carried out in the England and Wales, Chile and Norway, came to represent a standard, it is arguable that not all the steps were relevant to the conditions on the ground in most developing countries.

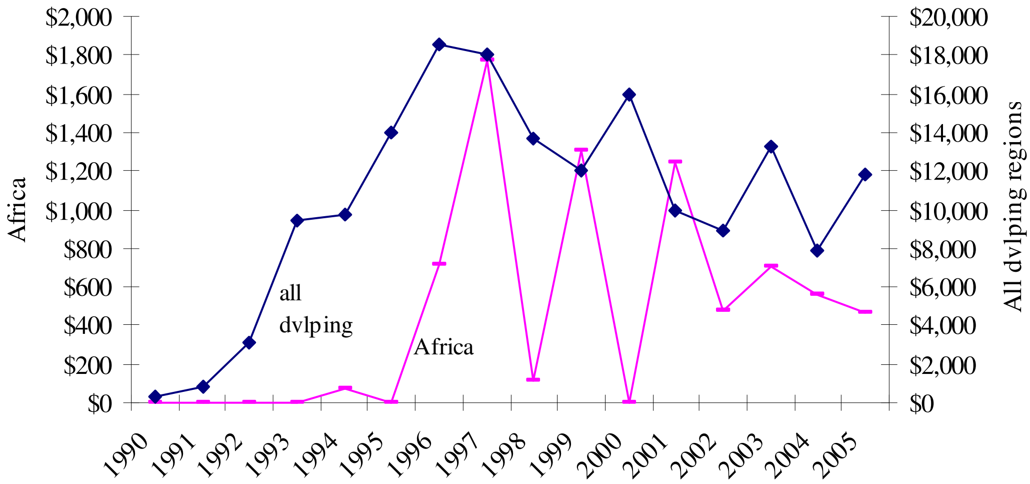
off its IPP development with a 210 megawatt (MW) combined cycle gas-fired plant undertaken by Saur International and Electricité de France (EDF). Five years later, the country would build West Africa's largest IPP, Azito, at 330 MW.[3] Egypt also became a magnet for private sector investment, with InterGen and EDF winning tenders to build approximately 2000 MW of power. Ghana, Kenya, Morocco, Tanzania and Tunisia, among others, also opened their doors to foreign and local investors. In 1997, later seen as the peak of investment, there was a record US$1.8 billion in IPPs in Africa (World Bank 2006b).

Although IPPs were considered part of a larger power sector reform programme, reforms were not far reaching. In most cases, state utilities remained vertically integrated and maintained a dominant share of the generation market, with private power invited only on the margin of the sector.[4] Policy frameworks and regulatory regimes, necessary to maintain a competitive environment, were limited. International competitive bids (ICB) for those IPPs that were developed were often not conducted due to tight timeframes, resulting in limited competition *for* the market and, due to long-term PPAs, no competition *in* the market. These long-term PPAs and often government guarantees and security arrangements, such as escrows and liquidity facilities, exposed countries to significant exchange rate risks. Finally, while Africa has seen private participation in greenfield electricity projects continue, private investments have not achieved the levels of the late 1990s, with 1997 representing the zenith as illustrated below.

[3] In terms of size, Azito has since been superseded by another West African plant, the 450 MW Okpai IPP in Nigeria, which came online in 2005. The largest IPP in Africa is, however, Morocco's Jorf Lasfar, a 1360 MW coal-fired plant, which will be discussed in section 3.3.2.

[4] Exceptions are Cote d'Ivoire, Morocco and Tanzania, where independent power projects contribute significantly (more than 50 per cent) to overall electricity production.

Figure 1: Greenfield IPPs in Africa vs. all developing regions (US$ million)



Source: based on authors compilation and Private Participation in Infrastructure Database (World Bank 2006b)
Note: Africa figures include only greenfield electricity projects in North Africa and Sub-Saharan Africa whereas world figures include both greenfield electricity and natural gas projects (however electricity projects account for 84 per cent of the total value).

Several factors explain the recent trends in investment. Private sector firms were deeply affected by the Asian and subsequent Latin American financial crises. The Enron collapse and its aftershocks also featured prominently in influencing American and European-based firms to reduce risk exposure in emerging and developing country markets and refocus on core activities at home. Futhermore, DFIs began to reconsider their position of restricted infrastructure investment, which had predominated throughout the 1990s. As concessionary funding became available again, many countries opted for publicly funded projects, rather than their private sector counterpart, such as Egypt, which has seen its current five year power investment implemented by the incumbent, state-owned utility, and supported entirely by concessionary loans.

Despite this revival of concessionary lending for power projects, investments are insufficient to address Africa's power needs, with only 25 per cent of the population presently with electricity access, and poor supply the rule, not the exception. With the original drivers for market reform still present, involvement of the private sector appears to be an inevitable part of the future.

This paper[5] seeks to evaluate IPPs in Africa by focussing on development and investment outcomes, namely the extent to which reliable and affordable power has been provided for the host country, and satisfactory returns on investments and new investment opportunities have been achieved.[6] [7] Case studies of eight African countries (Egypt, Tunisia, Morocco, Cote d'Ivoire, Ghana, Kenya, Tanzania and Nigeria), which have among the most extensive experience with IPPs, provide the empirical data for this analysis. At the heart of the analysis is a discussion of how the balancing of development and investment outcomes actually helps improve the sustainability of projects for public and private stakeholders alike. Contributing elements to success are also identified as the building blocks for more sustainable investments.

2. Africa overview

2.1 Economic and social development

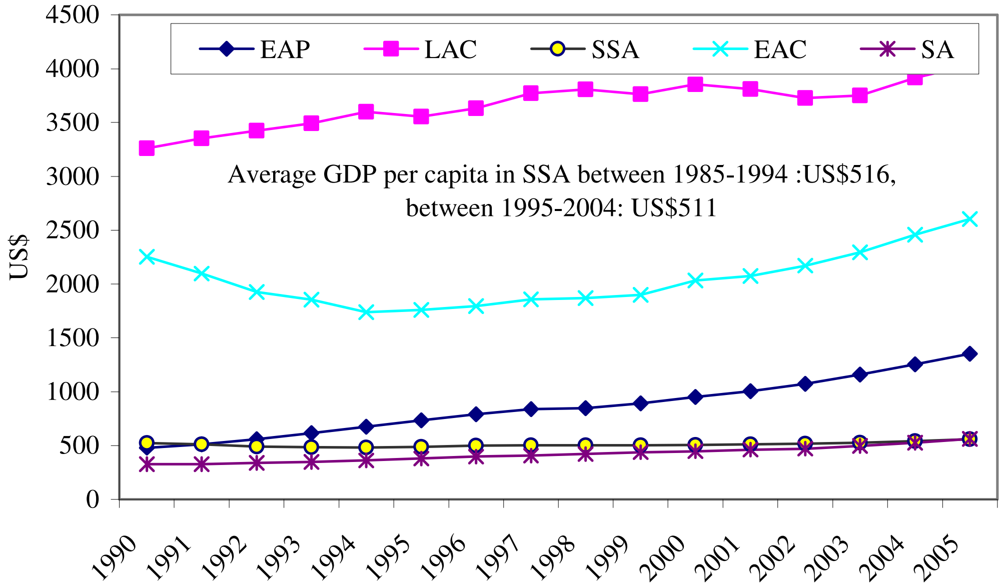
While there is extraordinary diversity across the 53 countries (and nearly 900 million people, representing about 14 per cent of the world's population) that make up the African continent, in terms of assessing the economic and social development status, the continent may be roughly grouped into North Africa and Sub-Saharan Africa (SSA). Furthermore, it is often useful to separate the relatively more developed country of South Africa (RSA) from SSA.

Figure 2: GDP per capita across developing regions 1990-2005 (in constant 2000 US$)

[5] The authors of this paper also contributed findings on IPP experiences in Egypt, Kenya and Tanzania to a global IPP study, led by Stanford University's Program on Energy and Sustainable Development (PESD). The overarching purpose of the PESD IPP study was to evaluate the IPP experiences across a number of countries and projects and thereby glean best and better practices for the future. See http://pesd.stanford.edu/ipps for information on PESD IPP study. It should also be noted that detailed research reports have been prepared as part of a broader assessment of African IPPs undertaken by the Management Programme for Infrastructure Reform & Regulation (MIR) that went beyond those studies included in the Stanford PESD analysis. The authors of this paper have also been co-authors of studies on Morocco (Malgas, Gratwick et al. 2007a) and Tunisia (Malgas, Gratwick et al. 2007b), as well as forthcoming studies on Cote d'Ivoire (Malgas, Gratwick et al. 2007c) and Ghana (Malgas, Gratwick et al. 2007d). In addition, an as of yet unpublished survey of Nigerian IPPs was conducted by the authors of this paper in collaboration with researchers at the Centre for Energy Research and Development at Obafemi Awolowo University in Ile-Ife, Nigeria. Further information on the MIR IPP research may be found at http://www.gsb.uct.ac.za/gsbwebb/default.asp?intpagenr=309.

[6] Neither the use nor the definition of development outcomes and investment outcomes is unique to this paper. What is, however, unique is the in depth analysis of such outcomes within the context of African IPPs. References to development and investment outcomes include Manibog (2003:22-25) and Victor (2004:11-13).

[7] Undoubtedly, there is overlap between the development and investment outcome, particularly when it comes to local investors who may be considered to be directly related to both outcomes.



Source: compiled by authors, based on World Development Indicators (WDI) (World Bank 2007a)
Notes: GDP: Gross domestic product; EAP: East Asia & Pacific; LAC: Latin America and Caribbean; SSA: Sub-Saharan Africa; ECA: Europe & Central Asia; SA: South Asia. North Africa has been excluded due to the fact that data is only available after 1999. It should be noted, however, between 1999 and 2005, average per capita GDP for North Africa was US$2,000 as reported in WDI.

The gross national income (GNI) per capita for North Africa (at US$1,700)[8] falls into the broader category of low and middle income countries and is slightly better than the average for East Asia and Pacific (World Bank 2005a:24; World Bank 2005b:33). A considerable contrast is seen in data for SSA (excluding RSA),[9] which records the lowest GNI per capita of all regions in the world at an average of US$342 (World Bank 2005a:24). Furthermore, unlike any other region, GNI per capita in SSA (again, excluding RSA) has been falling, as population growth has outpaced economic growth (World Bank 2005b:33).[10] Not only has the number of extremely poor doubled since 1981 to 313 million in Sub-Saharan Africa, the relative economic condition has also deteriorated as evidenced by the following: since 1981,

[8] Data is based on the World Bank's Atlas method, which employs a three year moving average conversion factor, adjusted for price, to reduce fluctuations in the exchange rate.
[9] The data for South Africa will be presented independently for a number of indicators. Although demonstrating extreme income inequality, RSA boasts an average per capita income of US$2,900 (equivalent to US$10, 130 in terms of purchasing power parity), which exceeds the average of middle income countries (World Bank 2005a:24; World Bank 2005b:33). Unless otherwise specified in the text, however, figures for Sub-Saharan Africa include RSA.
[10] Between 1975 and 1984, average annual GNI was reported as US$420; this dropped to US$354 between 1985 and 1994 and finally to US$309 between 1995 and 2003 (World Bank 2005b:33).

the average daily income of those people subsisting on less than one US dollar per day has dropped from 64 US cents to 60 US cents (World Bank 2005a:3-4).[11]

The trends highlighted above are reflected across a number of social development indicators. North Africa has seen its infant mortality rates drop by more than half to 31 per 1000 births between 1990 and 2003 (World Bank 2005b:313). In contrast, although infant mortality has fallen in SSA since 1990, it is still the developing region with the highest such rates (at just over 100 deaths per 1,000 births) (World Bank 2005a:11; World Bank 2005b:313).[12] Sub-Saharan Africa also records the highest mortality rates of all developing regions for children under five years of age, at 171 deaths per 1,000 children (World Bank 2005a:28).[13] Undernourishment is yet another revealing indicator, with SSA recognized as the developing region with the highest incidence as well as the only such region to see rates rise since 1990 (to well over 30 per cent of the population) (World Bank 2005a:5). In addition, although minor progress may be noted, Sub-Saharan Africa, together with South Asia, rank among those developing regions with the most limited access to improved sanitation facilities, with such access for just over one third of the population (World Bank 2005a:32; World Bank 2005b:317).

In accessing these and other economic and social development indicators against the Millennium Development Goals (MDG),[14] Sub-Saharan Africa, in contrast to all other developing regions, appears to have the fewest number of countries that have already achieved the MDG poverty reduction target (viz. halving the proportion of people living on less than US$1 dollar) (World Bank 2006a:3). Also noteworthy is the region of SSA's status with regard to the second MDG goal of ensuring primary school for all children by 2015; here, Sub-Saharan Africa records the greatest percentage of countries of all developing regions that are "seriously off track" in achieving the goal (World Bank 2006a:4-5). For the remaining six MDGs, SSA lags behind, including in its record with disease; malaria,

[11] In contrast, throughout the rest of the developing world, the average daily income of those living on less than one dollar per day has risen from 72 US cents to 83 US cents (World Bank 2005a:4).

[12] South Asia is second to Sub-Saharan Africa in terms of infant mortality, but records just over 70 deaths per 1,000 births (World Bank 2005a:11).

[13] Although a separate case was made for South Africa's GNI per capita, in terms of infant mortality, the country has not scored as well. It is among the few African countries to see infant mortality rise between 1990 and 2003 (from 45 to 53) (World Bank 2005b:313). One additional point should be made in this context, although HIV/AIDS is the cause of nearly 2.5 million deaths in Sub-Saharan Africa, since 1990 (more than in any other developing region), it is malaria that is the "leading killer" in Africa, with nearly a million people, mostly children, dying as a result of malaria *each year* in Sub-Saharan Africa (World Bank 2006a:12-13).

[14] The MDGs, among the most frequently used metrics for charting economic and social development progress, outline eight goals to be achieved between 1990 and 2015 related to improvements in the following areas: poverty and hunger; universal primary education; gender equality and female empowerment; child mortality; maternal health; the spread of disease (HIV/Aids and malaria); environmental sustainability and enhancing a global partnership for development (United Nations 2005).

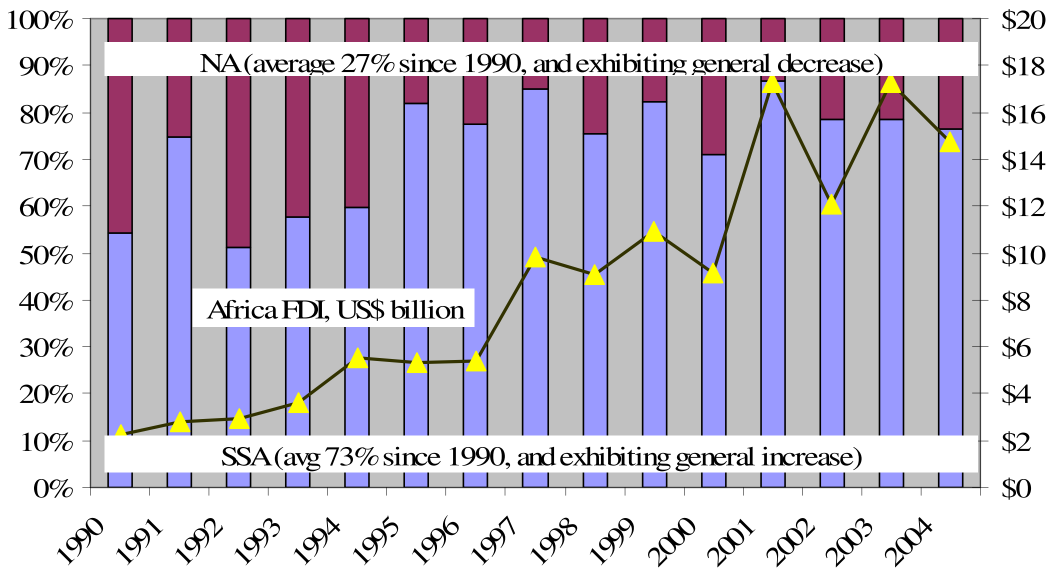
HIV/AIDS, and tuberculosis are all on the rise rather than nearing the goal of halting the spread of such epidemics (World Bank 2006a:12-13).

Thus, relative and absolute poverty continue to undermine social and economic development on the continent (with notable exceptions across North Africa and in other pockets including parts of South Africa), despite significant efforts to the contrary. Although an in-depth evaluation of such poverty is outside the realm of this paper, a basic understanding is crucial for two reasons, which represent a conundrum of sorts. On the one hand, the investment needs are huge.[15] On the other hand, there is limited public funding including concessionary loans, and foreign, private investors are often wary to put their money in projects given the level of underdevelopment and associated risk.

2.2 FDI and the investment climate

What is the record of investment and why? At first glance, the Figure immediately below, which records African foreign direct investment (FDI), including a sustained increase for Sub-Saharan Africa, does not appear to square with the discussion above, namely foreign private investors are wary to tread on African soil.

Figure 3: Africa Foreign Direct Investment



Source: Based on WDI (World Bank 2007a)

[15] Estache puts the total expenditure across all infrastructure (electricity, telecommunications, roads, rail, water and sanitation) to be 9 per cent of GDP between 2005 and 2015 to meet the MDGs (Estache 2005:15). Fay and Yepes in estimating investment needs across infrastructure between 2005 and 2010 (not specifically with MDG targets in mind) came to a figure of US$25,912 million or nearly 6 per cent of GDP (Fay and Yepes 2003:11).

While growth may be noted for the continent, it is important to recognize that FDI in Africa averages just 2 per cent of world totals for FDI between 1990 and 2004 (World Bank 2007a), despite the continent having 14 per cent of the world's population and 20 per cent of its land mass. The need for additional funding in virtually every sector—from health to electric power generation—is acute, however, the investment climate keeps many would-be financiers at bay. Although a wide range of indicators seek to report on the investment climate in Africa, this section briefly profiles just one of the main indicators. Not only is coverage limited of many of the major indicators,[16] a detailed discussion of such facts is also beyond the scope of this paper.

2.3 Credit ratings

Among the most widespread metrics for judging a country's investment climate are international credit ratings, which assess the likelihood that an entity (sovereign, bank or corporate) will repay its foreign currency or local currency financial obligations in a timely manner. Both short and long-term, foreign currency and local currency ratings are internationally comparable assessments. The leading rating agencies, Moody's Investors Service, Standard & Poor's & Fitch, assign ratings when a bond is first issued, and that rating helps determine how high the bond's interest rate will be. If the agencies assign a high rating, that means there is little risk of default, so the issuer can obtain a lower interest rate.[17]
It is the paucity of investment grade ratings that is the most striking feature of Africa. In North Africa, only Tunisia presently has an investment grade rating (Chambers 2007; *Sovereign Ratings Summary May 2* 2007). Throughout Sub-Saharan Africa, four countries have attained such ranking: Botswana, Mauritius, Namibia and South Africa. Morocco and Egypt, are, however, as noted above, considered to have relatively better investment climates and record just one notch below investment grade (BB+), followed by Lesotho and Nigeria (BB-). In addition, 15 other Sub-Saharan African countries have received a speculative grade rating from at least one of the major rating agencies, namely: Benin, Burkina Faso,

[16] Consider for example the Investment Climate Surveys as reported in WDI (World Bank 2006a:270-272). Senior managers are asked to report on the extent to which policy uncertainty, corruption, courts, crime, regulation and tax administration, finance, electricity and labour are considered major constraints to doing business. Only 13 countries throughout North Africa and Sub-Saharan Africa are included, and notably absent are Cote d'Ivoire, Ghana and Tunisia, which form part of this paper's analysis.

[17] Each of the major agencies differs slightly in their ranking of long-term credit ratings (short-term are not provided below):
Moody's ratings: from highest to lowest. **Investment grade:** Aaa, Aa1, Aa2, Aa3, A1, A2, A3, Baa1, Baa2, Baa3. **Speculative grade:** Ba1, Ba2, Ba3, B1, B2, B3, Caa1, Caa2, Caa3, Ca, C1.
S&P's ratings: from highest to lowest. **Investment grade:** AAA, AA+, AA, AA-, A+, A, A-, BBB+, BBB, BBB-. **Speculative grade:** BB+, BB, BB-, B+, B, B-, CCC+, CCC, CCC-, CC, D.
Fitch's ratings (same as S&P): from highest to lowest. **Investment grade:** AAA, AA+, AA, AA-, A+, A, A-, BBB+, BBB, BBB-. **Speculative grade:** BB+, BB, BB-, B+, B, B-, CCC+, CCC, CCC-, CC, D.

Cameroon, Cape Verde, The Gambia, Ghana, Kenya, Madagascar, Malawi, Mali, Mozambique, Rwanda, Senegal, Seychelles, and Uganda.

It should be noted that there is progress to report on the ratings front, considering the fact that in the last two years, five Sub-Saharan African countries (Kenya, Namibia, Nigeria, Rwanda and the Seychelles) have all received ratings from the major agencies, for the first time. However, the persistent dearth of investment grade ratings remains a potential obstacle to foreign investment.

2.4 African power sector

2.4.1 Power generation and resource endowments

Despite the widespread poverty documented above, which is interlinked with FDI and the investment climate, the continent is endowed with significant natural resources, many of which fuel its electric power generation.[18] This section briefly sketches Africa's coal, natural gas, hydropower and oil resources, together with electric power production and consumption patterns.[19]

Africa generated approximately 550 Terawatt hours (TWh) in 2005 or approximately 3 per cent of the world's total electricity generation (BP 2006:39). More than 40 per cent of this was generated in South Africa (BP 2006:39). The bulk of the remaining production was generated in North Africa.

It is coal that contributes most significantly to Africa's power generation or about 50 per cent of total electric generation—70 per cent for Sub-Saharan Africa and approximately 7 per cent across North Africa (World Bank 2005a:164; World Bank 2007a).[20] Explaining the disparity in usage is the fact that coal is found primarily in South Africa, which accounts for just over 5 per cent of total world coal reserves and the vast majority of coal found in both Africa and the Middle East (BP 2006:28).[21] [22] Nearly mirroring reserves, actual production

[18] It should be acknowledged that the resource curse or paradox of plenty as mapped by Terry Lynn Karl would not see poverty and resources at odds but rather a direct correlation between the two. For a detailed discussion see Karl and Gary (Karl and Gary 2004).

[19] Nuclear power production is minimal with South Africa's Koeberg station producing the only nuclear power on the continent; total consumption accounts for approximately 0.5 per cent of world nuclear consumption (United States Energy Information Agency 2005; BP 2006:33).

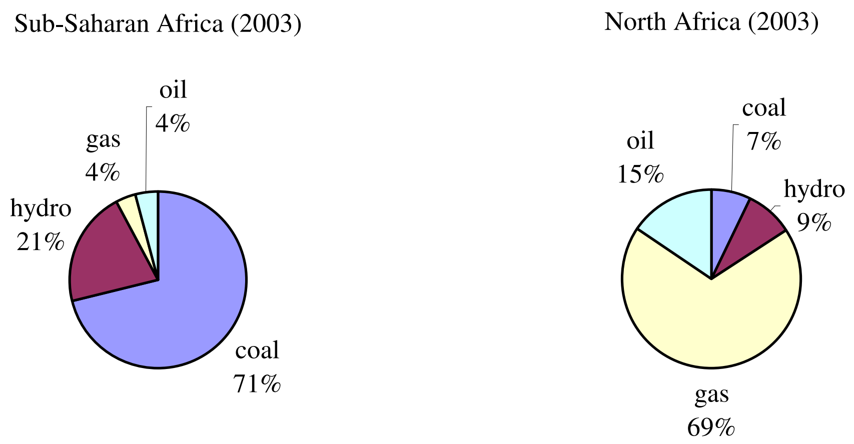
[20] An important exception to this breakdown in regional resources is the coal-fired Jorf Lasfar IPP, which increased Morocco's base load capacity by roughly 50 per cent.

[21] As defined in BP's statistical review: this refers to proved reserves of coal (or other resources), namely "those quantities that geological and engineering information indicates with reasonable certainty can be recovered in the future from known deposits under existing economic and operating conditions," (BP 2006:28).

[22] Zimbabwe contains approximately 0.1 per cent of the world's total proven coal reserves and 'other Africa' approximately 0.1 per cent as well (BP 2006:28).

of coal from Africa amounts to just under 5 per cent of world totals, again with South Africa accounting for 4.8 per cent of the total (BP 2006:30).[23]

Figure 4: Contribution to electric generation by fuel source, Sub-Saharan Africa and North Africa



Source: based on WDI (2007c)

Natural gas ranks second in terms of electric power generation for the continent at approximately 25 per cent of total electricity generated; as with coal, a wide disparity may be seen with present usage, with natural gas accounting for about 70 per cent of total generation for North Africa, and just under 5 per cent for SSA (World Bank 2006a:164; World Bank 2007a). Since 1990, North Africa has witnessed growth of more than 200 per cent in terms of electricity generated by natural gas (World Bank 2006b). Natural gas reserves are concentrated primarily in North and West Africa, explaining the disparity in usage, namely: Nigeria (5.2 Trillion cubic feet, Tcf), Algeria (4.6 Tcf), Egypt (1.9 Tcf) and Libya (1.5 Tcf). Significant natural gas discoveries have, however, been made in Southern Africa, with reserves of 4.5 Tcf, 2.2 Tcf and 2.0 Tcf, identified in Mozambique, Namibia and Angola, respectively.[24] Altogether, African gas reserves amount to approximately 15 Tcf or about 8 per cent of total, world proven reserves (BP 2006:17; Energy Information Agency 2006; Energy Information Agency 2007). Natural gas production does not mimic the reserve profile exactly, with top producers in descending order: Algeria, Egypt, Nigeria and Libya, which together, with the rest of Africa combined, account for 6 per cent of the total of world

[23] As of 2002, South Africa was the world's third largest net coal exporter (United States Energy Information Agency 2005).
[24] Although comparatively less at 800 Billion cubic feet (Bcf), discoveries of natural gas are slowly changing the face of electric power generation in Tanzania. Also of significance in terms of natural gas development in Southern Africa is the Mozambique gas pipeline, bringing gas from the Temane and Pande fields in Mozambique to Sasol (in South Africa) along a 537 mile long pipeline, since 2004.

gas production at the end of 2005 (BP 2006:19). Important to note that, in contrast to coal, the continent consumes less than half of what it produces (accounting for only 2.6 per cent of the world's natural gas consumption) led by consumption in Egypt, Algeria and South Africa, which imports from Mozambique (BP 2006:22). Much of the consumption profile could, however, change with the growth of natural gas-fired power plants in Nigeria as well as the development of the West African Gas Pipeline Project (WAGP), which will transport Nigerian natural gas to Benin, Togo and Ghana, and which could eventually be extended to Cote d'Ivoire and Senegal. The Kudu Gas Project, which involves harnessing off-shore gas for a 800 Megawatt (MW) power plant, and which like the WAGP has been in the planning stages for several decades, could also, bring more gas into the Southern African consumption profile. Smaller projects like Tanzania's Songo Songo Gas-to-Electricity project have been instrumental in changing the demand profiles of individual countries.

Hydropower is presently contributing nearly 20 per cent of electric generation for the continent—21 per cent in Sub-Saharan Africa and about 10 per cent in North Africa (World Bank 2005a:164; World Bank 2007a). It is estimated that 95 per cent of Africa's technical hydropower potential remains *un*exploited (Aardt 2006). The technically exploitable capability is estimated at 1917 TWh/year or approximately just less than 15 per cent of the world's total technically exploitable capability (World Energy Council 2001; World Energy Council 2004:209). Installed capacity, at just over 20,000 MW, presently accounts for just less than per cent of total world installed capacity, despite the abundance of the resource (World Energy Council 2001; World Energy Council 2004:214), which points to low demand profile as well as the investment gap.[25] In terms of hydropower that is presently exploited, it has been harnessed across Sub-Saharan Africa, with some of the largest installations (after Egypt) in: Democratic Republic of Congo, Mozambique, Nigeria, Zambia and Ghana (World Energy Council 2001). Furthermore, much of the technically exploitable capability cited above is located across these and other Sub-Saharan African states that lack natural gas, oil and coal resources. It should be noted that a number of the IPPs reviewed in detail in this paper were developed with the immediate cause of alleviating electricity shortages due to drought (in largely hydro-based systems). Thus, reliance on hydropower goes a long way in explaining IPP developments across the continent. It should also be noted in this context that although the continent has not yet seen any large-scale hydro-powered IPPs, among the latest IPPs planned is Uganda's Bujagali, a 250 MW hydroelectric plant, with lead sponsor formerly the American-based AES and now Industrial Promotion Services (IPS), a subsidiary of Aga Khan's Fund for Economic Development (AKFED).

[25] Consumption also amounts to approximately 3 per cent of the world's total (BP 2006:35), with limited amounts of power generated via hydropower through North African linkages with the Middle East, through an interconnector with Jordan.

Finally, this discussion ends with oil, which amounts to about 8 per cent in terms of electricity generation across Africa. North Africa has seen electricity generated by oil drop since 1990 (by 20 per cent) as natural gas becomes increasingly ubiquitous. Meanwhile, although only amounting to 4 per cent of total generation in SSA, it has grown by more than 150 per cent since 1990 (World Bank 2006a:164; World Bank 2007a). Proven oil reserves, totaling approximately 115 thousand million barrels (as of the end of 2005), are concentrated in North and West Africa, primarily in Libya (40 thousand million barrels) and Nigeria (35 thousand million barrels), followed by Algeria, Angola, Sudan and Egypt (BP 2006:2).[26] Actual production paints a similar picture, with the top producers presently, in descending order: Nigeria, Algeria, Libya, Angola and Egypt (BP 2006:2,4). All told, however, as of end-2005, Africa's proven reserves are only about 10 per cent of world proven reserves and production accounts for approximately 12 per cent of total world production, with the continent as a whole, consuming just 3.5 per cent of total world production, led by consumption in Egypt, South Africa and Algeria (BP 2006:4,6).[27]

2.4.2 Installed capacity and consumption patterns

National electricity systems in African countries are small by international standards (with the exception of South Africa). Installed capacity and per capita consumption are among the lowest across developing regions (World Bank 2007a). South Africa has a total nominal capacity of 42,000 MW,[28] followed by Egypt with 22,500 MW, then both Nigeria and Morocco with 5,000 MW (Egyptian Electricity Holding Company 2004; International Atomic Energy Agency 2005; Eskom 2006).[29] Thereafter countries taper off with Mali possessing a mere 280 MW and Burkina Faso, 120 MW (International Atomic Energy Agency 2005). One important, more recent development has been the addition of emergency, also termed temporary power, namely units that are rented generally for 1-2 years during a power shortage. Although most installations are generally no larger than 100 MW, given the

[26] Other smaller producers include (in decreasing scale): Gabon, Republic of Congo (Brazzaville); Equatorial Guinea; Chad and Tunisia (BP 2006: 2).
[27] Michael Klare describes the increasing foreign interest in Africa's oil and gas resources as well as increased development and military aid as well as arm sales (Klare 2001:217-221). See also Karl and Gary (2004), as previously referenced.
[28] The case of South Africa is noteworthy in this context. Although the country has had significant excess capacity for two decades, RSA is presently facing a tight supply/demand balance, which impacts on the rest of the Southern African region, which is interconnected through the Southern African Power Pool (SAPP) and to which Eskom is the primary supplier of power. Ten Southern African countries are linked through SAPP, namely South Africa, Botswana, Swaziland, Mozambique, Lesotho, Namibia, Zimbabwe, Zambia, Angola and Democratic Republic of Congo (DRC). A further two countries, Malawi and Tanzania, are part of SAPP, but not linked.
[29] Important to note, however, in the case of Nigeria, only about 3,000 MW of the approximately 5,000 MW installed are operational as 2007, due to age and maintenance of plants.

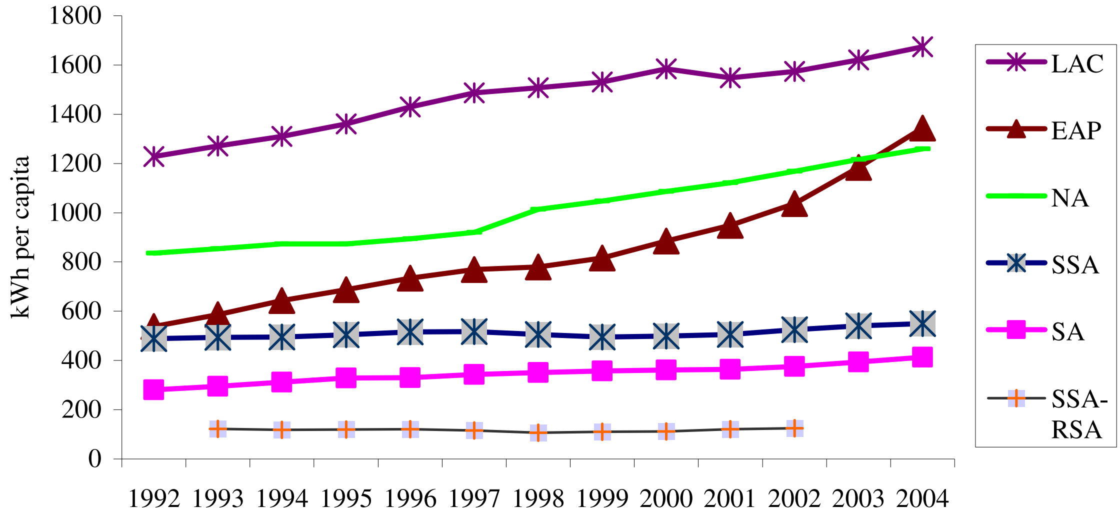
small base of installed capacity, emergency power, has come to play a significant role in African Electricity supply industries (ESI) in the past decade.[30] [31]

Electricity consumption averages approximately 554 kilowatt hour (kWh) per capita per year. This figure masks a wide disparity: 900 kWh for North Africa and 457 kWh for Sub-Saharan Africa, with the average for SSA falling to 124 kWh if South Africa is excluded (World Bank 2005b:245). For a global comparison, annual average per capita consumption across developing countries is 1,155 kWh (United Nations Development Programme 2006, table 22). This stands in sharp contrast to high income countries, which averaged 10,198 kWh (United Nations Development Programme 2006, table 22). Furthermore, if South Africa is excluded, SSA is the only region in which per capita consumption is actually falling. Due to its comparatively minute electricity markets, Africa has been easily passed over as a potential destination for IPP developers.

[30] Take for example Aggreko, one of the leading firms to provide such power in Africa, which after 10 years is active in providing power plants in about 10 countries across the continent, including about 40 per cent of Uganda's capacity as of 2006. It should be noted that if all forms of utility projects are considered, Aggreko's activity spans 21 countries across Africa.

[31] Although such power has been particularly critical in staving off the effects of reduced hydroelectric capacity as a result of drought, it is understandably more expensive due to the short duration of the contracts. In most countries, these extra costs are subsidized, with the cost of such subsidies in Uganda between 2005 and 2011 to be roughly estimated as equivalent to the total cost of the 250 MW Bujagali hydropower project (*Uganda* 2007:4).

Figure 5: Electricity consumption per capita (kWh) 1992-2004



Source: compiled by authors, based on World Bank (2005b; 2007a)
Note: Data for SSA-RSA only available between 1993 and 2002. Developing countries within the region of Europe and Central Asia (ECA) have been excluded from this graph with the average for ECA twice that of LAC.

In Sub-Saharan Africa, where 83 per cent of the African population lives, approximately one quarter of the population has access to electricity, and in some countries, as little as 5 per cent of the population has access to electricity. There are presently 438 million people living in rural areas without electricity (as opposed to 109 million without electricity in urban areas) (International Energy Agency 2006:156). Access rates in rural areas may be as low as 2 per cent, as seen in Tanzania (Marandu 2002). In Kenya, rural access rates are just 3.8 per cent (Kenya Ministry of Energy 2000). Electricity consumption is concentrated in urban areas among the middle and upper classes, larger business owners and the public sector (Estache 2005:33,40). While a quick survey may deem the 75 per cent unserved as 'latent demand', this demand is characterized by its poverty and often located great distances from any electricity infrastructure, i.e. it presents more of a public policy challenge than an investment opportunity.

2.4.3 Organization of the electricity sector

How are Africa's electricity supply industries structured and organized? African ESIs throughout both North and Sub-Saharan Africa have traditionally been organized as vertically integrated monopolies, with ownership in the hands of the state (Nellis 2005:9-11). Over the last two decades, power sector reform has been propagated throughout the continent, and there are signs of movement toward corporatization and commercialization (including via management contracts). IPPs have been introduced in about 15 countries, and regulators have been established in approximately 22 countries, with efforts underway to set up such agencies in 12 more countries (Eberhard 2007:15).[32] However, no African ESI has yet to be fully unbundled (vertically and horizontally), and nowhere is there evidence of either wholesale or retail competition (Besant-Jones 2006:22; SAD-ELEC 2006:9).[33] [34]

3. Private participation in the power sector

3.1 International trends

Although power sector reform is not very advanced in Africa (if measured by the extent of unbundling and the introduction of competition), there is nevertheless considerable evidence for private sector participation, prompted by the necessity of plugging the investment gap.

The World Bank's private participation in infrastructure (PPI) Database, among the most comprehensive sources for tracking private sector investment over the last two decades, provides a picture of such participation and investments (World Bank 2007b).[35] To contextualize within the broader framework of infrastructure reforms, private participation in energy (defined as electricity as well as natural gas, for which electricity represents 84 per cent of the actual total value of projects) has ranked second to telecommunications in terms of investment flows, followed by transportation and water. According to PPI, between 1990 and 2005, energy investments with private sector participation have occurred in 105 developing countries, through 1,307 projects, totalling US$298 billion. Latin America and the Caribbean account for the largest number of projects (460), and the largest chunk of investment (US$124 billion), and Sub-Saharan Africa the least (with 69 projects valuing US$7 billion).[36]

[32] Many of the newly created regulators have, however, come under fire for what have been perceived as inconsistent and often arbitrary rulings, thereby increasing rather than decreasing overall sector risk, (Eberhard 2007:1).

[33] SAPP includes a small short-term energy market (also known as STEM) and a bilateral market. Volumes of electricity traded on STEM reached a high of 842 GWh in 2002 and since then have declined, recording a mere 178 GWh in 2005, attributed mainly to the failure of Mozambique and Zambia participating due to rehabilitation and maintenance projects (Southern African Power Pool 2006: 14). SAPP's bilateral contracts amounted to just over 10 TWh in 2002, and have been increasing steadily to 16 TWh in 2004 and 19 TWh in 2005 (Southern African Power Pool 2006:14-15).

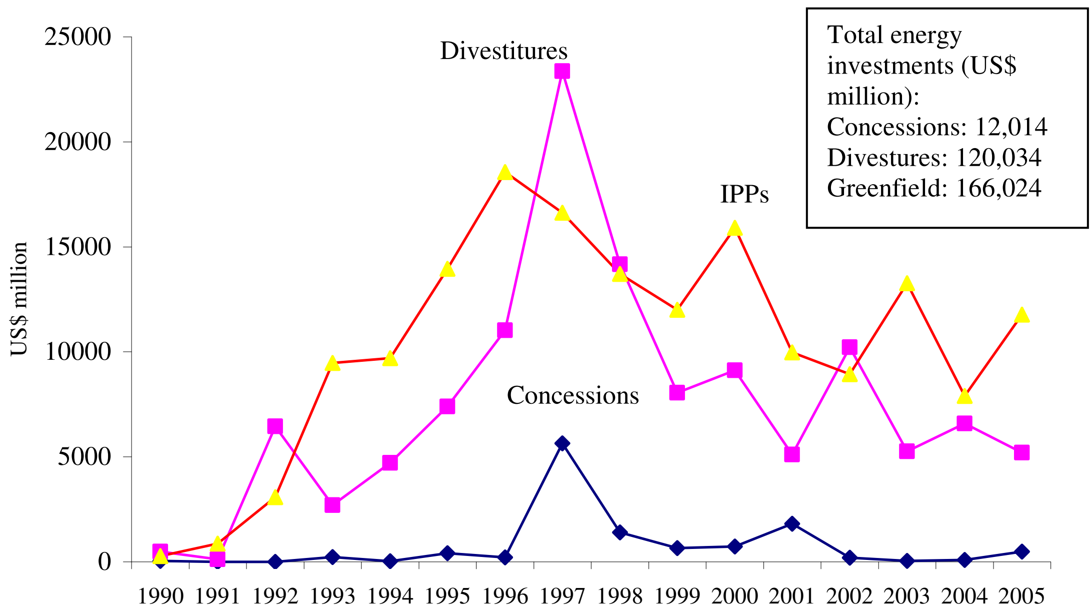
[34] It should be noted, however, that evidence for partial unbundling is found in Kenya and Uganda, where generation has been separated from transmission and distribution. Nigeria is also in the process of unbundling (Besant-Jones 2006:22). Furthermore, in Namibia and South Africa, local government is involved in distribution, and in Nigeria, the Power Holding Company of Nigeria (PHCN) has been nominally unbundled, however, PHCN still maintains a coordinating role.

[35] More comprehensive reviews of results are reported in Kessides (2004) and Besant-Jones (2006: 30).

[36] Detailed analysis of PPI in electricity is provided in Sader (1999), Jamasb (2002:3-12) and Kessides (2004:167-180).

Depicted immediately below is a breakdown of projects by type of investment. Important to note is that although divestitures represented an unprecedented US$23,370 million in 1997 (due mainly to several divestitures of utilities in Latin America), by far the most significant investment over the course of the past 15 years has been in greenfield investments, which largely represent investments in IPPs, for a total of US$166,024 million. To put this figure in context, in the 1990s, IPPs represented about one tenth of private FDI in developing countries (Victor, Heller et al. 2004:2).

Figure 6: Private sector participation in energy



Source: based on World Bank (2007b) and authors' compilation
Note: In addition, management and lease contracts have led to increased private sector participation; however, the volume is small at just 215 contracts over the 15 year period.

The year 1997 is cited as the peak year for private participation in electricity (Sader 1999:15; Covindassamy, Oda et al. 2005:1,4). Thereafter, the East Asia and Pacific financial crises followed by the Latin American financial crisis together with the collapse of Enron and its aftermath caused a sharp decline in private investment across developing regions (Covindassamy, Oda et al. 2005:6; Woodhouse 2006a:135).[37] Thus, by the end of the decade, new investments in IPPs (to take one example of PPI) were fewer and further between. In 1997, over 90 IPPs reached financial closure, followed by less than 50 in 1998 and approximately 50 in each subsequent year (World Bank 2006b). In addition, approximately 30 of the 35 IPP contracts in the Philippines were renegotiated (albeit mutually renegotiated),

[37] The Enron aftermath referenced above was characterized by a retreat of American and European-based investors back to their home markets.

and most contracts were also renegotiated in Thailand, Malaysia and China. There is also evidence of refusal to honour contracts in China and India, as well as cancellation of contracts in Indonesia and India (Woodhouse 2006a).

Despite widespread financial crises and several high profile contract cancellations, a recent study reports that remarkably few projects have actually undergone stress, defined as "electricity projects that have been terminated before their term; are under arbitration; are still ongoing, but declared unsatisfactory by either the investors, the host government, or the lenders; or went through a stress period, but were successfully worked out" (Covindassamy, Oda et al. 2005:19). Only 4 per cent of total PPI electricity projects are categorized as having experienced such stress. Furthermore, of all PPI electricity projects, IPPs tend to be the least susceptible—with only 3 per cent of IPPs noted as undergoing stress—which is credited to long-term contracts that are not subject to market fluctuations and the relative protection that projects enjoy from political machinations, as opposed to, for example, distribution projects (Covindassamy, Oda et al. 2005:61). Furthermore, also noteworthy, is the recent, albeit modest, increase in PPI electricity projects (Kerf and Izaguirre 2007:1).[38]

Notwithstanding the rise and fall of PPI in electricity, PPI activity has been substantial. The question, however, is how substantial? As previously indicated, the value of the deals with private participation in energy between 1990-2005 was about US$300 billion, with IPPs amounting to about 55% of this total (World Bank 2007b). Contrast this figure with the International Energy Agency's estimate for new investment needed to meet the projected increase in electricity demand and to replace old infrastructure between 2002 and 2030, in developing countries, namely US$5.2 trillion (International Energy Agency 2004:191). While seemingly substantial, the investment flows in the larger scheme have actually been quite minimal.

While the figures, cited above, may be intrinsically interesting, of critical importance is how and why these projects and investment flows translated into change and the nature of the change that occurred.

[38] The Stanford PESD IPP study, led by Erik Woodhouse, concluded that 21 or 34 projects were renegotiated. The discrepancy in findings between the Stanford PESD IPP study and Covindassamy's Projects Under Stress may be explained by several factors. First, Covindassamy's work, based primarily on the PPI database, included planned projects, cancelled projects and projects under construction, whereas the Stanford PESD work considered only existing IPPs. Second, the PPI database also includes captive plants that occasionally sell power to the grid, which are outside the scope of the Stanford PESD work. Finally, although 21 of 34 projects were renegotiated, Stanford PESD records only 4 out of 34 projects as having had negative outcomes, which although still larger than Covindassamy's figure, are more comparable with the definition of stress employed in the World Bank study.

3.2 IPPs in Africa: an overview

Approximately 40 IPPs have been developed in Africa to date (depicted in Figure 1 below). With few exceptions, they represent only a fraction of total generation capacity and have mostly complemented incumbent state-owned utilities.[39] [40] Nevertheless, IPPs have been an important source of new investment in the power sector in a dozen African counties, eight of these countries (Cote d'Ivoire, Egypt, Ghana, Kenya, Morocco, Nigeria, Tanzania and Tunisia) account for three-quarters of installed IPP capacity and about 70 per cent of all IPP investment.

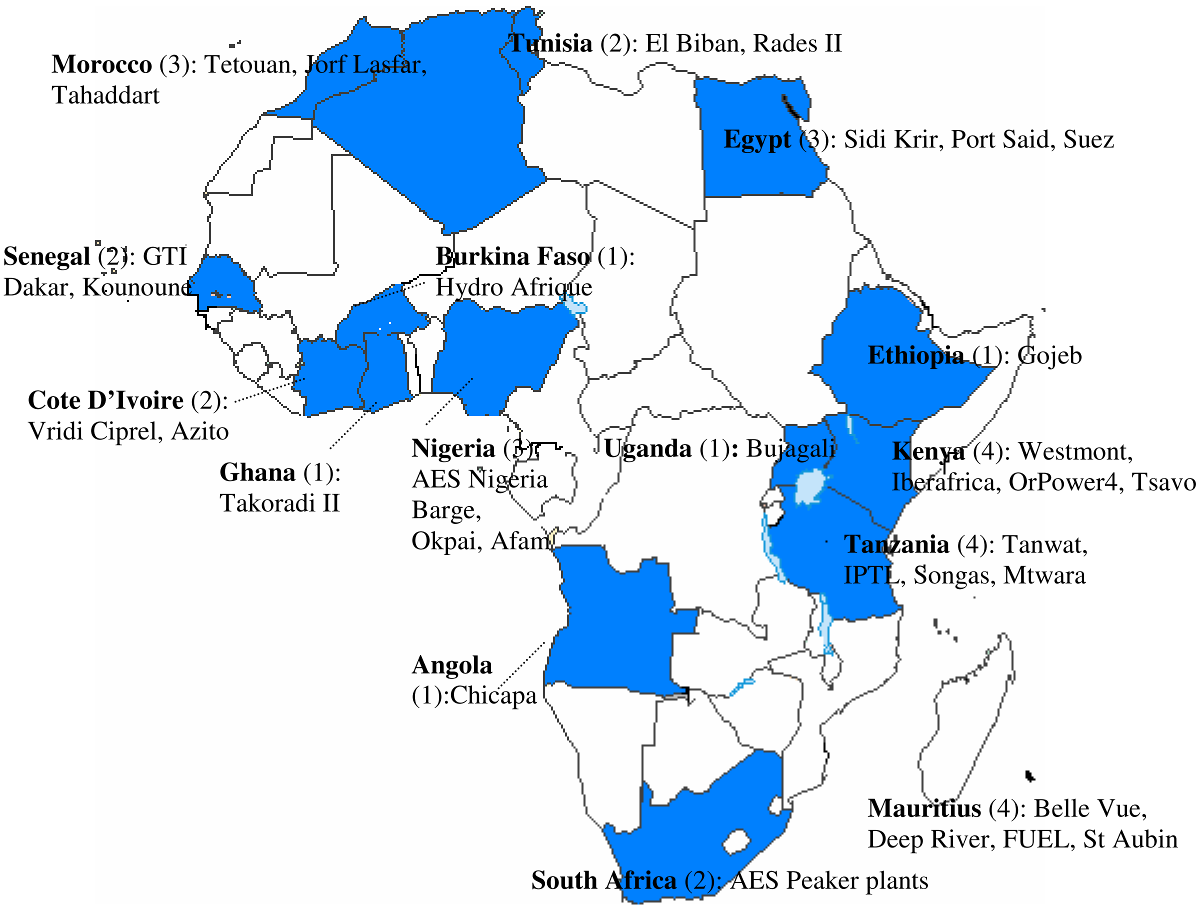
How and why did IPPs take root at all? There was a general trend of power sector reform, driven in part by the DFIs, which played a particularly large role in Africa. There was also a growing sense within countries that the private sector could inject much needed cash and energy into the electricity sector. Thus, states solicited IPPs and, as will be discussed below, offered them terms and conditions to try to mitigate some of the risks outlined above. This section briefly explores the impetus for IPP development as well as how projects have unfolded across the different regions, with a special focus on the experiences of Egypt, Morocco, Tunisia, Kenya, Tanzania, Cote d'Ivoire, Ghana and Nigeria.[41] A list of the IPPs in these countries is provided in Table 1.

[39] The International Energy Agency reports installed capacity for Africa at approximately 112,000 MW as of 2004; with IPP installed capacity roughly equal to 9,500 MW, IPPs are just less than 9 per cent of total installed capacity on the continent (International Energy Agency 2006:527). It should, however, be noted that the IEA total installed capacity figure appears to be inflated given data on systems (as presented in section 2.4.2); with no additional source providing comprehensive data on African installed capacity, the authors have included the reference nonetheless.

[40] Exceptions are Cote d'Ivoire, Tanzania and Morocco where IPPs presently contribute more than half of all generation.

[41] Although Mauritius has four IPPs (which, at approximately 200 MW combined, account for about 37 per cent of installed capacity and a little less than 25 per cent of production, as of end-2005), the country has not been included in this paper's sample, primarily due to the idiosyncrasy of the contracts. The IPPs, which are all cogeneration plants, provide power and steam to the country's sugar mills throughout the crop season, reducing their contribution to the state-owned utility by about 30 per cent. During this time, the shortfall in production is made up by seven Continuous Power Producers (CPPs), privately owned by the sugar mills. With installed capacity of 40 MW, roughly equal to the IPP shortfall, the CPPs also have long-term take or pay contracts with the state (2007b).

Figure 7: Greenfield IPPs in Africa



Source: based on the World Bank (2007b) and authors' compilation

Notes: Figure includes all IPPs for SSA and North Africa that have reached financial closure (with the exception of the South African AES IPPs where financial closure is expected by December 2007). Not included are IPPs that were cancelled, namely Congo's Songa IPP, Ghana's SIIF Accra and Osagyefo Barge (also known as Western Power). Kenya's Westmont has since concluded its 7-year contract. Construction has yet to be started on South Africa's peaker plants and Uganda's Bujagali.

Table 1: African IPP sample, general project specifications[42]

Country/ Project	Size (MW)	Cost (US$ million)	Fuel/cycle	Contract type	Contract Yrs	Project tender -COD
Egypt						
Sidi Krir	683	413.9	Natgas/steam cycle	BOOT	20	1996-2002
Port Said	683	340	Natgas/steam cycle	BOOT	20	1998-2002
Suez	683	338	Natgas /steam cycle	BOOT	20	1998-2003
Morocco						
Jorf Lasfar	680+680[43]	1500[44]	Coal	BTO	30	1994-2000
CED	50	58.5[45]	Wind	BTO	19	1995-2000
Tahaddart	384	364.9	Natgas/combined cycle	BTO	20	1999-2005
Tunisia						
Rades II	471	260.7	Natgas/combined cycle	BOO	20	1997-2002
SEEB	27	30[46]	Natgas/open cycle	BOO	20	2000-2003
Kenya						
Westmont	46	35	Kerosene/gas condensate/gas Turbine (barge-mounted)	BOO	7	1996-1997
Iberafrica	56	65	HFO/medium speed diesel engine	BOO	7, 15	1996-1997
OrPower4	13	54	Geothermal	BOO	20	1996-2000
Tsavo	75	85	HFO/medium speed diesel engine	BOO	20	1995-2001
Tanzania[47]						
IPTL	100	120	HFO/medium speed diesel engine	BOO	20	1997-1998
Songas	180	316	Natgas/open cycle	BOO	20	1994—2004
Mtwara	12	8.2	Natgas/gas-fired reciprocating engine	BOO	2[48]	1994-2007
Cote d'Ivoire						
CIPREL	210	105.6[49]	Natgas/open cycle	BOOT	19	1993-1995
Azito	330[50]	233	Natgas/open cycle	BOOT	24	1996-2000
Ghana						
Takoradi II	220[51]	110	Light crude oil/single cycle	BOOT	25	1998-2000
Nigeria[52]						
AES Barge	270	240	Natgas/open cycle (barge-mounted)	BOO	20 (2 parts)	1999-2001
Okpai	450	462	Natgas/combined cycle	BOO	20	2001-2005

[42] See Appendix A for detailed specifications on each project.
[43] The first 680 MW refer to an existing facility; the greenfield investment was an additional 680 MW.
[44] Project costs included upgrade of the coal receiving facility and infrastructure for transporting coal to ONE's Mohamedia plant.
[45] Project costs quoted in Euros at €45.7, converted to US$ on August 24 2006 (€1=US$1.28).
[46] Project costs of SEEB, Songas (Tanzania) and Okpai (Nigeria) include the gas infrastructure.
[47] Tanzania's Tanwat (depicted on African IPP map), is a 2.5 MW facility, selling excess power into the grid and therefore is not included in this chart.
[48] A long-term PPA for 20 years may be signed with TANESCO.
[49] Investment cost €87.8m or 57.6 billion CFA, with the average 1994 conversion of US$ to CFA, 545.100.
[50] The initial project concept included specifications to raise capacity to 420 MW.
[51] The initial project concept included specifications to add a second phase of 110 MW and convert to combined cycle, however, lack of funding has limited the completion of this phase.
[52] In addition, Nigeria expects to bring Afam V and VI online in 2007.

Notes: COD: commercial operation date; natgas: natural gas; BOOT: Build Own Operate Transfer; BTO: build, transfer, operate; BOO: Build Own Operate; HFO: heavy fuel oil.

3.3 North Africa: Egypt, Morocco and Tunisia

A distinguishing feature of the North African experience is its investment climate which is significantly better than the rest of Sub-Saharan Africa (comparable with conditions in emerging economies in Latin America and South East Asia) and which inevitably impacted risk ratings, the cost of capital and IPP contract terms. Also of significance with the North African plants is that they were generally tendered under international competitive bids (ICB), and that plants were largely gas-fired with two important exceptions in Morocco.

3.3.1 Egypt: from 15 to 3[53]

Unable to finance plants from its own coffers and faced with less interest by development finance institutions in the 1990s to provide finance for infrastructure, the Government of Egypt (GoE) turned to the private sector for new generation capacity. It passed the requisite legislation in 1996 to allow for private ownership of generation assets. The following year, new investments provisions, including sovereign guarantees, a five year corporate tax exemption, currency conversion, repatriation of profits, and protection against nationalization and expropriation, were also adopted to create a climate that would attract investors. The Egyptian Electricity Agency, later reorganized as Egyptian Electricity Holding Company (EEHC),[54] was given the mandate to drive the IPP process forward.

The IPP plan advertised to investors included a fleet of 15 projects, which would be phased in over several years. The first tender was for a 682.5 MW gas-fired steam cycle plant, under a 20 year BOOTcontract, with EEHC as the guaranteed off-taker, backed by the Central Bank of Egypt. Fuel was to be provided by the state-owned Egyptian gas firm (and the cost would be passed through to the Egyptian Electricity Transmission Company (EETC). Interest was high, and GoE ultimately negotiated among the lowest tariffs for developing country IPPs at 2.54 US cents per kilowatt hour (kWh).[55] The second and third IPPs, 683 MW each and both gas-fired steam cycle, followed along nearly the same lines, with the major distinguishing feature of the latter two being that they were unable to secure domestic, US dollar denominated debt. Project finance, with limited recourse to parent companies, was arranged for all three deals.

[53] This section is based on Eberhard and Gratwick (2007)

[54] Egyptian Electricity Agency (EEA) was reorganized as EEHC in 2000, through Law 164. The Egyptian Electricity Transmission Company (EETC) pays capacity charges to IPPs on behalf of EEHC.

[55] Fifty one firms applied for pre-qualification, 21 were initially pre-qualified, with 11 selected to receive the tender, and 9 actually bid.

With 12 IPPs outstanding and considerable interest among the present IPP sponsors, namely InterGen, Edison and Electricité de France (EDF), Egypt met an unexpected development that would ultimately change the course of private generation in the sector. At the end of 2002, the Egyptian pound lost nearly half of its value, which had the effect of doubling the local price of IPP charges (which were denominated in US dollars). Despite pressure to change contract terms, the three IPPs (Sidi Krir, Port Said and Suez) remained unaffected. What did, however, change was the future builds. The 12 outstanding plants were cancelled, and the government, instead, with the help of development finance institutions, embarked on a publicly funded expansion. To date, the only new private power is being supplied by independent service providers (ISP), which offer both generation and distribution services to captive customers (primarily tourist resorts), but amount to only 1 per cent of installed capacity. A final development in Egypt's short IPP history is that, although not necessarily linked to the cancellation of the IPP programme, equity in all three IPPs has since changed hands, with InterGen and Edison selling their shares to Globeleq by mid-2005, and EDF selling its shares to Kuasa Power, a fully owned subsidiary of Tanjong Energy of Malaysia by early 2006. As Globeleq repositions itself as a developer of greenfield plants, it too has put its equity stake up for sale, with a conditional sale agreement completed in the second quarter of 2007. Tanjong Energy is also the leading shareholder of the potential buyer (with Tanjong holding 55% and a Saudi Arabian firm, Al Jomaih, holding the balance).[56] Much remains to be seen, but Tanjong has indicated that if policy frameworks are established, the firm might even consider venturing into bilateral contracts (without government guarantees or any long-term PPA with EEHC), which along with the country's ISPs, might move it one step closer to achieving wholesale competition.

3.3.2 Morocco: a local innovation[57]

Much of the same pressure, related to public funding constraints, seen in Egypt, was also at play in Morocco. The main difference in Morocco was the magnitude of the pressure: in the mid 1980's Morocco's public debt amounted to 110 per cent of GDP (reduced to 60 per cent by the mid-1990s).

[56] It should be noted that, with the exception of Sub-Saharan Africa, Globeleq has put up its entire portfolio of assets for sale, as the firm seeks to reposition itself in developing new greenfield plants (rather than in taking over existing plant, as it did for the first few years since it was founded) (*Africa: Globeleq re-thinks asset sale after disappointing offers* 2007:7). Although a surprise to many outsiders, this development was always part of the firm's strategy, as Globeleq sought to plug the hole left by the exodus of many private investors following the downturn in the global power market in 2001-2002. While there has been much discussion of Globeleq's sale of its worldwide portfolio, a decision has since been made by the firm not to sell of any of its Sub-Saharan assets due to the fact that "indicative bids received did not meet the value placed on those businesses by Globeleq and its shareholders," (*Africa still just a flicker in the global M&A market* 2007; Globeleq per com 2007).

[57] This section is based on Malgas, Gratwick and Eberhard (2007a).

In addition to helping to reduce the country's debt burden, the introduction of IPPs was expected to: free up funds for increased electricity access programmes, help meet growing electricity demand, contribute to diversification of fuel supply,[58] assist in the benchmarking of state-owned plants, and finally bring about both efficiency gains and a reduction in tariffs. Although these aims were common among procurers of IPP power, Morocco's actual procurement plan was radical. Through its first IPP, Jorf Lasfar, a 1360 MW coal-fired plant, Morocco increased its base load capacity by roughly 50 per cent.[59]

Like Egypt, Morocco created a policy environment that clearly communicated its willingness to private investors. Legislation was passed to allow for concessions in the electricity sector in 1994 (albeit following a Build, Transfer, Operate model to conform with existing Moroccan law). Equally important were the suite of investment incentives that were made available. As seen in Egypt, the project company was exempt from corporate tax for the first five years (followed by a 50 per cent reduction in years six through 10). Customs duties and value added tax (VAT) exemptions were also extended. A sovereign guarantee was provided for early termination and foreign exchange risk for Jorf Lasfar and was backed by a World Bank guarantee as well. An international competitive bid was commenced by the government in October of 1994 to build two additional units at Jorf Lasfar and operate the entire facility under a concessionary agreement for a period of 30 years. Three different consortia submitted proposals: (1) Asea Brown Boveri (ABB) and Consumer Michigan Services (CMS); (2) ADS and General Electric (GE); and (3) Alstom, with the first consortium ultimately selected.[60]

Similar incentives would be extended for Morocco's second and third IPPs (excluding the World Bank guarantee), which like Jorf Lasfar were record setters. Compagnie Eolienne de Detroit (CED), the second IPP, was the first privately funded wind project in Africa, which would contribute 50 MW to the grid; Energie Electrique de Tahaddart (EET), the third IPP, was the country's first gas-fired combined cycled plant, at 384 MW, with debt provided entirely by domestic sources. The resulting cost of power is seen as attractive, as confirmed by public and private stakeholders, however the actual tariff for IPPs remains confidential, with the one exception of CED, at €3.9-5.9 cents per kWh (German Ministry for Economic

[58] Significant headway had been made in terms of diversification since Morocco's independence in 1956 when the country relied on hydropower for nearly 90 per cent of its generation. At the inception of IPP development in the country, coal and oil were the dominant fuel, followed by hydropower.

[59] Jorf Lasfar was a two part project: a brownfield (680 MW) and a greenfield (680 MW). In two other countries in Africa, IPPs have accounted for the majority of electricity production. Cote d'Ivoire is also noteworthy for its capacity increase through IPP investments, with IPPs representing more than half of the country's installed capacity and production. Finally, although amounting to only one third of total installed capacity in Tanzania, Songas and IPTL, the country's two IPPs did, with recent droughts, contribute more than 50 per cent of total generation.

[60] It is important to note, however, that both CMS and ABB have (since 2007) sold their equity to the Abu Dhabi National Energy Company, also known as TAQA.

Cooperation and Development Report 2002). Finally, it is worth noting that all three projects, like those in Egypt, were structured as limited recourse project finance deals.

Although the three IPPs are seen as success stories, untroubled by any macroeconomic shocks to date,[61] state involvement in the power sector remains (as originally envisaged by the government at the inception of the IPP programme). The government, through the state-owned utility (ONE), took a sizeable stake in the latest IPP Tahaddart (with nearly 50 per cent of the equity share) as it wanted to be involved in influencing the project and financing arrangements and thereby ensuring the best possible deal.[62] The next two wind farms, amounting to a total of about 400 MW, are expected to be owned and operated by the public sector. ONE is also launching a 472 MW hybrid power project, with 20 MW fuelled by solar power and 452MW by natural gas. Furthermore, feasibility studies are also being conducted by the state to explore the development of nuclear power. Amidst this state-led activity, there is, however, one IPP, for which competition is proving tough. Twenty firms have pre-qualified as of early 2007 to develop by 2011 a 1,320 MW coal-fired BOOT with a long-term (30 year) PPA with ONE. Thus Morocco's story is successful, but it does not clearly illustrate complete acceptance of private generation as the way forward. In fact, the success of the proposed new hybrid market, where large customers will choose their supplier and smaller customers will buy at a fixed tariff, depends on an optimal mix of IPPs and state-owned generation. Finally, it is worth noting that the drivers for IPPs are not as strong as at the beginning of the IPP programme as Morocco's external debt situation has ameliorated significantly and access rates have improved.

3.3.3 Tunisia: investment successes and conundrums[63]

Of the three North African countries, Tunisia may be said to have had the best investment profile at the start of its IPP programme (with investment grade ratings with S&P, Moody's and Fitch), which had a clear impact on its ability to attract bids and cement favourable deals from the perspective of the host country. It is, however, also the only country in the North African pool of cases that ran a selective bid (in addition to one international competitive bid) process.

With economic liberalization (rather than unavailability of public funds) the primary motivating force, Tunisia enacted the requisite legislation providing for private participation

[61] Morocco has been unaffected by macroeconomic shock, however, it has also had a more proactive monetary policy to ward off the effects of macroeconomic shock with the dirham pegged to a basket of currencies dominated by the Euro, which has had a stabilising effect on repayments due to the exchange rate remaining relatively stable since the first IPP was developed.

[62] Initially ONE was a minority stakeholder and only took 20 per cent of the equity. When one of the original sponsors, EDF, left, the state utility assumed a bigger stake (namely, 48 per cent).

[63] This section is based on Malgas, Gratwick and Eberhard (2007b), with, as indicated previously, the authors of this paper, co-authors of the piece.

in generation in 1996. This paved the way for an ICB organized by the state utility, Societé Tunisienne d'Electricité et du Gaz (STEG), in 1997 for a 471 MW gas-fired combined cycle plant. Seventeen firms responded to the RFP, from which a consortium made up of the Public Service Enterprise Group (PSEG), Marubeni and Sithe was selected, for providing the lowest price per MW.[64] As seen in both Egypt and Morocco, incentives for the sponsor included a five year tax holiday. Also provided to sponsors of Rades II, the project company, was an exemption on VAT and customs duties for all imported equipment that could not be sourced locally prior to COD, as well as a government commitment for the completion of permit applications.

Similar to most of the Moroccan plants, investors in Rades II did not require a government guarantee, a fact which undoubtedly reflects on the sound investment climate. Also noteworthy was how Tunisia structured the capacity charges. STEG negotiated that payments be tied to a basket of currencies (roughly 60/40 split between Euros and US dollars) to mitigate the impact of future devaluations.[65] The payment negotiation was a direct response to earlier experiences in the 1980s, when the state utility financed another plant (Rades A) with loans denominated in Japanese Yen, which increased its value three-fold relative to the dinar and led to a major cash crunch for STEG.

It should be noted that neither the currency denomination nor the lack of guarantee biased the outcomes for Tunisia's IPP. The country was able to secure an investment cost for Rades II of US$554 per kW installed, which is comparable to the price of IPPs in Egypt, despite the fact that the combined cycle gas turbine (CCGT) technology is more costly to procure.

Tunisia's second IPP has not been nearly as smooth in its development. Unlike Rades II, the second plant, also known as Societe d'Electricite d'El Biban (SEEB) was an outgrowth of a government policy, enshrined in the 1999 Hydrocarbons law, to attract foreign companies to explore oil and gas. Under this framework, the SEEB plant (a 27 MW open cycle gas-fired unit) was selectively tendered, as mentioned previously, in contrast to all other IPPs discussed thus far in North Africa, with CME serving as the initial project sponsor, together with Centurion Energy and Caterpillar Power Ventures. Project incentives included the five year tax holiday, characteristic in the project pool, which facilitated in reducing the debt payback period to five years. Problems began to arise in 2004, a year after COD. Gas for the plant was contaminated, which had the effect of damaging SEEB's ejectors and hence adversely impacting the plant's performance. Approximately one year later, water entered

[64] Sithe exited the project during the development phase, and the shareholding was subsequently split 60/40 between PSEG and Marubeni.

[65] Apart from STEG's efforts at minimising foreign currency risks, the Central Bank of Tunisia has made preserving the value of the currency with respect to its major trading partners a major priority. The Tunisian dinar has devalued only by an average of 3.5 per cent per year with respect to the US$ over the 10 year period from 1993-2002, making it one of the most stable currencies on the continent of Africa.

the gas well, and gas could no longer be extracted. Captive to one gas well, SEEB has been out of operation from August 2005 until the present. During this time, no money has exchanged hands between STEG and SEEB.

The SEEB plant is not alone in its challenge to secure fuel quality and security of supply. In 2003, BG Group and the Government of Tunisia entered into a Memorandum of Understanding (MoU) for the development of the Barca project, a 500 MW combined cycle gas plant. One year later, however, negotiations with BG stopped due to concerns about the security of gas supply for the project. STEG has since been charged with developing the plant, albeit with a different fuel sourcing arrangement. A greater success story for IPPs lies with Tunisia's wind power developments; about 100 MW of privately constructed wind power is expected to come online in 2007, followed by another 200 MW between 2008 and 2011. Finally, although contingent on an interconnection with Europe, STEG has indicated that it will invite tenders for El Haouariam, a 1,200 MW CCGT IPP in 2007, with approximately two-thirds the capacity earmarked for Italy. Although Tunisia appears to be active in pursuing IPPs, like Morocco, state-led talks to develop nuclear power for electricity are also underway, with security of supply cited as the reasons behind both nuclear power and state involvement.

3.4 East Africa: Kenya and Tanzania

There are striking contrasts between the experiences in North Africa and those in East Africa, mostly related to the investment climate. But there are also notable similarities such as the push for fuel diversification and the host of investor incentives.

3.4.1 Kenya: two different waves[66]

Kenya stands out in this sample as the only country against which there was an aid embargo at the inception of IPPs. Therefore, while in other countries development agencies had signalled their unwillingness to fund infrastructure projects, in the case of Kenya there was a general no- funding policy across all sectors until the country enacted a series of political reforms. This embargo had the effect of sending the signal to potential investors that Kenya was *not* a favourable place to do business.

Nonetheless, investors did come, shortly after the passage in 1996 of legislation opening the sector to private firms, and Kenya represents among the earliest African countries to have IPPs reach commercial operation. Investors were offered custom exemptions and tax holidays until commissioning, similar to other FDI in Kenya, together with full repatriation of profits. In 1997, Westmont, a gas turbine plant burning kerosene and gas condensate, and

[66] This section is based on Eberhard and Gratwick (2005)

Iberafrica, a medium speed diesel engine plant burning HFO (both under 50 MW) began providing power to the majority state-owned transmission and distribution utility, Kenya Power and Light Company (KPLC). Although less costly than the emergency power that the state would procure several years later to reduce the impact of drought, this first wave of IPP power did not come cheap.[67] The investment climate alone is not to blame, however, with selective tendering (rather than open international competitive bidding) followed and prices alleged to have been inflated by corruption as well.

Lessons from this first wave were quickly learned. The second round of IPPs (which comprised a 12 MW geothermal plant, OrPower4, and a 75 MW medium speed diesel engine plant, Tsavo) involved international competitive bidding, and contracts were ultimately approved by the newly established independent regulator. Kenya's Electricity Regulatory Board (ERB)[68] was also instrumental in helping to reduce tariffs negotiated in a subsequent PPA between Iberafrica and KPLC. Although it remains unclear whether the same outcome could have been achieved had Kenya's regulator been sited within a ministry, the independent status of ERB has helped to make the body's actions ultimately less subject to controversy as the regulator appears to be at arms length from investor and host country government alike, i.e. both major stakeholders in the IPPs.

This shift between the first set of IPPs and the second, however, has not meant that it has been entirely smooth sailing. Compared to hydroelectric plants, owned and operated by Kenya Generating Company Limited (KenGen), the state-owned generator, IPPs appear expensive, and with persistent drought and increased reliance on IPPs, the second wave of IPPs have come under public pressure for capacity charge reductions as well. The International Finance Corporation (IFC), although part of the aid embargo during the 1990s, has since played a critical role in terms of helping the Tsavo project company, in which it holds equity, to resist pressure to renegotiate contract terms. Also of significance has been the role of the development-minded IPS and local partner KPLC Pension Fund in helping to mitigate local pressure for Tsavo and Iberafrica, respectively.

Kenya's generation future appears to have both more public and private generation investments in store: OrPower4 is in the process of adding a further 36 MW (after several years of negotiations, as well as an investigation by MIGA, and a reduction in the tariff by OrPower4 for this second phase); 100 MW in emergency power has been procured under a 12 month contract with Aggreko; a new 70 MW gas turbine will be built by KenGen, and a new diesel generator of 80 MW is expected from the private sector. Both the projects have,

[67] Iberafrica, which has been benchmarked against IPPs in developing countries of similar size and technology proves to be among the most costly at US$1,161 per kilowatt installed.

[68] It should be noted that in July 2007, ERB was transformed into the Energy Regulatory Commission, which regulates the entire energy sector (including petroleum).

however, encountered delays for diverse reasons, including, in the case of the 80 MW IPP, the final award being disputed by one of the bidders in the ICB.

3.4.2 Tanzania: overcapacity and under-capacity[69]

Although no aid embargo existed in Tanzania at the time of its first IPPs, the country was still suffering from a poor investment climate. Emerging from a command economy, there was little explicit protection against nationalization, the currency was not convertible, and firms were unable to repatriate profits.[70] In the country cases summarized until now, there has been a repeated emphasis on the dwindling role of finance provided by DFIs at the inception of the IPP developments.[71] In the case of Tanzania, however, a slightly different model is in evidence with the World Bank, the Swedish International Development Cooperation Agency (Sida), and the European Investment Bank (EIB) playing an important role in facilitating these new investments, given the abovementioned conditions.[72]

Private sector involvement was sought for the development of Tanzania's gas reserves together with a power expansion at Ubungo and gas sales to third parties in the early 1990s, on the heels of legislation opening up the sector to private firms. This project, known as Songo Songo gas-to-electricity, was advertised to 16 firms and ultimately undertaken by a consortium led by TransCanada.[73] It was expected that the gas-fired open cycle 60 MW expansion would be up and running within less than a year, despite the hefty infrastructure development (not to mention financial support) required. By 1995, however, with work on Songo Songo outstanding, a second deal was struck by government for 100 MW of diesel engines, known as Independent Power Tanzania Limited (IPTL), under a 20 year PPA.

At the time, Tanzania could absorb power from one plant, but certainly not two. Why then was the second deal reached? The impetus for IPTL may be attributed to a host of factors, including alleged corruption. Paramount among these factors is poor power sector planning and coordination. What ensued in the aftermath of the IPTL deal was a lengthy attempt at cancellation and renegotiation of this second plant as well as temporary postponement of

[69] This section is based on Gratwick, Ghanadan and Eberhard (2006).

[70] The following terms would, however, eventually be extended to investors in the power sector: a five year tax holiday, customs exemptions, full repatriation of profits, protection against nationalization and currency conversion via a pass-through to bulk purchasers of power in the following month.

[71] Technical assistance was often still made available, and there are cases where lenders did syndicate loans, take equity, and even provide debt.

[72] Several small self-producers also sell power to TANESCO, including Tanwat (2.5 MW), Kiwira Coal Mine (6.0 MW)—which were the first two IPPs---and Kilombero Sugar (2.5 MW). They sell only small amounts of excess power, and thus are not treated within the scope of this paper.

[73] Efforts to develop Tanzania's Mnazi gas field also occurred during this time, with Tullow Oil selected to develop the field in 1994. Tullow would, however, ultimately opt out, citing poor economics, and was replaced by the Artumas Group, which brought a 12 MW plant online in 2007, powered by Mnazi gas.

Songo Songo, led by the World Bank. Thus by the end of the 1990s, rather than having two IPPs in operation, the country had none.

IPTL would eventually emerge in 2001 from an international arbitration settlement, at the World Bank's International Centre for Settlement of Investment Disputes, with nearly US$25 million shaved off its investment costs, representing a reduction of about 15 per cent. By that time, demand in Tanzania had picked up and capacity from both IPTL and Songo Songo were needed. The two plants, which reached COD in 2002 and 2004, respectively, would soon help the country to avert load shedding, but at a steep cost, as seen in Kenya as well. In 2005, costs for the IPPs absorbed 69 per cent of Tanzania Electric Supply Company Limited's (TANESCO), the state-owned utility, total revenue, while contributing 45 per cent of the total power generated. Of these, IPTL's charges weighed more heavily: the plant represented 37 per cent of total IPP generation but 62 per cent of total IPP charges. Although seemingly inconceivable, in 2006, IPP production amounted to 55 per cent of generation, but charges amounted to 96 per cent of TANESCO's revenues, due to a significant drop in TANESCO's revenues as well as increases in the price of HFO (Ghanadan and Eberhard 2007:33).

Presently the government is stepping in to help bail out the ailing utility. Plans for future IPPs include 200 MW from Kiwira Coal & Power, to be fuelled by indigenous coal, with units originally due online in December 2006, June 2007 and December 2007—however, with Kiwira unable to raise finance for the 135 kilometer (km) transmission line, the project is presently delayed. One hundred MW from Richmond Development Corporation, which had no prior experience in developing power project, slated for September 2006, was delayed as well, although 80 MW is finally up and running after the Government of Tanzania provided funds to transport generators. This project has since been sold to Dowans Holdings, based in the United Arab Emirates. Neither of these projects, nor developments related to the Mnazi Bay Gas to Electricity project (presently supplying 12MW) was competitively procured. Meanwhile, since the end of 2006, 40 MW of emergency power have been commissioned from a subsidiary of Alstom and another 40 MW from Aggreko.

3.5 West Africa: Cote d'Ivoire, Ghana and Nigeria

In West Africa, two of the countries included in this overview (Cote d'Ivoire and Nigeria) have experienced civil strife, however, with different impacts on their IPPs. Ghana has been free of civil strife, but plagued by persistent drought, and several parallels may be seen with Kenya and Tanzania, as described above.

3.5.1 Cote d'Ivoire[74]

The story of Cote d'Ivoire, where IPPs provide more than half of the country's generating capacity, is noteworthy for a number of reasons. Compagnie Ivoirienne de Production d' Electricité (CIPREL), a 210 MW open cycle plant fired by domestically produced natural gas, was among the first IPPs to come online in Africa. With major shares held by SAUR Group and EDF, this IPP started producing power in 1994. At the time, Cote d'Ivoire's investment climate was considered among the best in the region, with GDP growth at 7.7 per cent per annum. This relatively favourable investment climate would help attract bids for the second IPP, Azito, during its international competitive bid in 1996.[75] Ultimately a consortium composed of Cinergy, ABB, IPS and the Commonwealth Development Corporation (CDC)[76] was selected to develop the plant, with the deal safeguarded by a sovereign guarantee (as well as a partial risk guarantee from the World Bank's International Development Association, IDA). In 2000, when Azito, a 330 MW gas-fired open cycle plant, came online, it represented the largest IPP in the West Africa (since surpassed by the Okpai plant in Nigeria). Both plants have been instrumental in helping the country to avert the consequences of drought in a largely hydro-denominated sector.

Within months of Azito's PPA being signed (and shortly before project completion), the country was subject to a political coup. Since 1999, lengthy periods of civil unrest have been punctuated only briefly by moments of peace.[77] During this time of civil unrest, revenues of the national utility, Compagnie Ivoirienne d'Electricite (CIE), have been reduced by approximately 15 per cent,[78] which has had a direct impact on the state's ability to invest in new, and much needed, electricity infrastructure. Although revenues have fallen, there has, however, been no impact on the IPPs—neither damage to the plants nor suspension of PPA payments has occurred.

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[74] The material for this section is based on a forthcoming survey led by Malgas on Cote d'Ivoire's IPP experience (2007c). The authors of this paper are also collaborators in the work.

[75] Also noteworthy is that, with the exception of Kenya's IPPs, CIPREL and Azito are the only projects in this survey that were not granted extended tax holidays, which may be due to a suite of different factors including the fact that the return on investment ultimately reflected the single largest investment incentive together with the fact that these were among the first IPPs with a set framework as of yet undeveloped.

[76] Most of CDC's equity shares in projects were replaced by Globeleq, which was spun off of the CDC Group in 2002 and was established to invest in power projects in Africa, the Americas and Asia. CDC also invested in Tanwat, Tanzania's first IPP, which has not been included in this analysis as it was largely a captive plant producing 2.5MW and only selling excess power to the grid. It should, however, be noted here that Globeleq remains entirely owned by CDC.

[77] As of 2007, peace appears to be here to stay, with a democratic election on the horizon (expected in October 2007).

[78] The 15 per cent reduction in revenues are due to civil unrest, with electricity customers in the north encouraged by the opposition group to default on payments (and, at the same time, the opposition threatening CIE of serious consequences should power supply be interrupted).

In addition, the country has seen a significant currency devaluation. Although planned, the devaluation occurred just after the signing of the first PPA with CIPREL, which could have, but ultimately did not, impact on the US dollar denominated charges.

One of the ways that CIE has been able to honour its IPP payments, despite the drop in revenues and currency devaluation, is through additional foreign exchange gained by the export of power to Benin, Burkina Faso, Ghana, Mali and Togo. Thus, even amidst the civil unrest, Cote d'Ivoire has continued to define itself as a major power hub for the region.

New generation capacity is now presently needed, to meet domestic demand as well as Cote d'Ivoire's growing foreign markets. As noted above, the state's coffers are not able to finance new investments due primarily to a drop in revenue during the period of civil unrest. Thus, the government[79] has turned to owners of Azito and CIPREL to sponsor the next new large investment. As of mid-2007, no deals have been arranged, but several points are worth reporting. First the sponsors of the first two IPPs and potentially the next one are no longer exactly those of the mid-1990s, with EDF having sold its shares in CIPREL (amounting to approximately 35 per cent)[80] to SAUR Group together with all smaller investors (amounting to approximately 12 per cent). Globeleq, which assumed CDC's shares in 2002 and among the investors in Azito, considered selling its shares in the plant, in line with its new strategy of concentrating on greenfield investments in 2006, as previously mentioned (in footnote 56), however, a decision has since been taken to hold on to the asset. In contrast to EDF's exit and Globeleq's attempted exit, IPS, has indicated its interest in developing the last phase of Azito. With Cote d'Ivoire's unprecedented past, it remains unclear what the next record-breaking development will be exactly.

3.5.2 Ghana[81]

Although not affected by civil unrest, Ghana's IPP story is a turbulent one. The country has just one operating IPP, Takoradi II, but there have been several attempts to launch additional projects. While all attempts have come to nought, due to the fact that such failures inevitably impact on the investment climate and the perception by stakeholders, they will be discussed briefly below.

[79] It should be noted that CIE has no mandate to contract new investment, which is undertaken by the government via the Ministry, due to the existing management contract. The utility entered into a concession contract in 1990 with the French firm Bouygues, which expired in 2005. A second contract has since been agreed upon until 2020. Although the Ministry has indicated that it intends drafting a plan for the way forward before the expiry of the current contract, there are no near-term plans to unbundle and/or privatize the sector.

[80] EDF has, however, remained involved in CIPREL in a technical management consulting capacity.

[81] The material for this section is based on a forthcoming survey led by Malgas on Ghana's IPP experience (2007d).

In 1998, amidst drought conditions, the Ghanaian National Petroleum Corporation (GNPC) spearheaded a project, which was intended to facilitate private participation in the ESI. GNPC bought a 125 MW barge-mounted generator from Italy's Ansaldo Energy, as part of a plan to harness recently discovered off-shore gas reserves. ICB procedures were not followed and the project, which has come to be known as Osagyefo Barge, has been shrouded in controversy. Although almost a decade has passed since the project was closed, not one kilowatt has been generated, and like Kenya's Westmont plant, it sits idle even as the country suffers from severe power cuts. The main stumbling block has been the fuel source: GNPC has been unable to convince either the Government of Ghana or any private investors to build the requisite gas infrastructure. Presently on the cards is a plan to bring the plant onshore to Effassu and tap into the West Africa Gas Pipeline, however, the site that has been earmarked for Osagyefo Barge is not connected to the pipeline. With no investors willing to fund a connection, prospects for the plant look bleak.

SIIF Accra, although initially billed as an emergency power project, with a 36 month contract with the government, was keen to establish a longer term presence in the country. Developers, Cummins and Wartsila, thus expected to extend their contract after the third year, i.e. SIIF Accra appears to straddle the definition between emergency power and IPP. The 39 MW steam-cycle diesel generator was negotiated during the drought of 1998. After the contract had been signed and with construction underway, drought conditions reversed due to heavy rainfall, making SIIF Accra's promised power unnecessary. When the plant reached completion at the end of 2000, the firm began to submit invoices, reflecting the agreed capacity charges, to the government. No payments were made, however. Subsequently, the new government that came to power in January 2001 responded by insisting that SIIF Accra should have discontinued construction once it was evident that the hydrological conditions had reversed. SIIF in turn continued to demand payment. Reconciliation was attempted, but the parties were not able to agree to a settlement. Meanwhile, Cummins and Wartsila shipped the 39 MW unit to Benin where it has provided power. Subsequently, SIIF sought international arbitration. As of mid-2007, there has been no resolution reached by the parties.[82]

There is, however, as mentioned at the outset, one standing, operating IPP (a 220 MW single cycle plant powered by light crude oil), which like those plants in Cote d'Ivoire and Tanzania has provided much needed power to Ghana's ESI. The plant was developed by CMS (90 per cent), with remaining equity held by Volta River Authority (VRA), the state

[82] There have, however, been a number of emergency power plants, which did successfully produce power—30 MW provided via an 18 month contract by Aggreko and 30 MW provided by Cummins for 24 months, starting from June and September 1998, respectively. Detailed treatment of such plants is beyond the scope of this paper.

utility.[83] Initially, it was intended that the plant would be built in two phases, first 220 MW single cycle, followed by 110 MW combined (steam) cycle. One of the stumbling blocks in developing this second phase, as cited by investors, has been the lack of government guarantees (which were granted in the first phase of Takoradi II). Government guarantees do not, however, appear to be forthcoming, and the government of Ghana has indicated that the engineering, procurement, and construction (EPC) costs are too high. There is a possibility that the recent change in ownership, with CMS selling its shares in 2007 to TAQA, the Abu Dhabi National Energy Company, may help jumpstart this pending second phase.

Meanwhile, several other investors, including CENPower, have offered to provide up to 360 MW of gas-fired generation to the country, as well as a consortium of mining firms (AngloGold Ashanti, Newmont Mining, Golden Star Resources and Gold Fields), whose businesses have been seriously impacted by persistent power cuts. The mining consortium's 80 MW plant is, as of mid-2007, feeding power into the grid, with the plant expected to be transferred to VRA later this year.

3.5.3 Nigeria[84]

Nigeria, the last country in this overview, is a fitting place to end the discussion as it is presently embarking on a full-scale overhaul of its power sector and therefore has the potential to learn the most from the previous countries profiled in this paper. The plans for power sector reform involve vertical and horizontal unbundling and selling off all of the state's generation and distribution assets (six generation companies and eleven distribution companies), previously the National Electric Power Authority (NEPA), which were reorganized under the Power Holding Company of Nigeria (PHCN) in 2005.

The recent flurry of activity is part of a longer history of reform, with legislation passed in 1998 allowing private participation in the generation sector. The first IPP, sponsored by Enron, emerged amidst the emergency power situation of 1999 when just over a third of the country's total capacity was in operation and load shedding was becoming increasingly widespread. Initially, stakeholders, namely Enron, NEPA and the Lagos State government, agreed to a 90 MW barge-mounted diesel plant to be run on liquid fuel, which would then be followed by a permanent facility comprising a 560 MW plant, using open cycle gas turbines (OCGT), both under a common PPA. International competitive bidding practices were

[83] Apart from the one IPP, Ghana's ESI is dominated by VRA, which owns all of the hydro generation and one thermal plant (Takoradi I). VRA also maintains a monopoly over transmission and distribution with the exception of the northern areas of the country, where the Electricity Company of Ghana (ECG), also a publicly-owned company, provides transmission and distribution services.

[84] The Nigeria material included in this paper is based on an as of yet unpublished survey of Nigerian IPPs, which was conducted by the authors of this paper in collaboration with researchers at the Centre for Energy Research and Development at Obafemi Awolowo University in Ile-Ife, Nigeria in 2006. Additional interviews have since been completed by the authors of this paper for 2007 data.

overlooked, and the deal was negotiated within months, with the expectation that the first plant would be online by December 1999.

Due to public pressure, however, this initial deal was modified, and renegotiations would carry on for twelve months through 2000. Major objections were raised: the lack of transparent and competitive bidding; the type of fuel to be used, the fact that the plant would not be penalized for poor performance; that the project company would receive excessive contract termination payments; and that payments could bankrupt the state and national utilities.

During the renegotiation, the original plan for the land-based 560 MW plant was shelved.[85] The major items to be renegotiated involved increasing the initial plant from 90 MW to 270 MW (9 units of 30 MW each) and changing the fuel from liquid fuel to natural gas, both of which had the effect of reducing the capacity charge to approximately US$19.00/kW/month, and a final investment cost of US$240 million. It was agreed that the capacity charge would be flat, for the 13.25 year duration of the contract, and that it would be indexed to the Consumer Price Index (CPI) of the Organization for Economic Cooperation and Development (OECD). Furthermore, capacity charges would be payable in dollars. There was to be no separate fuel supply agreement between the sponsor and the fuel supplier; instead, fuel was to be provided by NEPA/PHCN, which would contract it directly from the Nigerian Gas Company. As a final security measure, the PPA was to be backed by a sovereign guarantee. By the end of the renegotiations, ownership had changed hands, with AES acquiring 30 per cent of this facility in September 2000 and increasing its ownership to 95 per cent in December 2000. Yinka Folawiyo Power Limited (YFP), a local partner, purchased the remaining 5 per cent equity share of the plant**.**

Although agreement was reached and the plant has been online since 2001, PHCN has fallen behind in capacity payments (which are still perceived to be high by the public). In addition, the government has yet to issue the tax exemption certificate, with causes linked to the perceived high capacity charges. Presently, negotiations are underway between sponsors and the regulator for further reductions to the capacity charges and agreements related to overall performance. Yet another recent stumbling block has been the fuel supply, with civil unrest in the Niger Delta leading to damage to the pipeline supplying AES Barge Limited.

The second IPP, like that in Tunisia, grew out of an additional reform mandate—namely for oil companies operating in Nigeria to harness previously stranded (in this case flared) gas for power. Investment incentives extended to sponsors are the same as for all upstream gas projects and include protection against nationalization and expropriation, and full repatriation of profits. Also known as Okpai, this 300 MW combined cycle plant is operated

[85] The plan for the 560 MW plant was shelved by Enron, but not abandoned, with development expected to follow after the first phase of the project.

by Nigerian Agip Oil Company (NAOC), with majority ownership by Nigerian National Petroleum Corporation (60 per cent), followed by NAOC (20 per cent) and Phillips Oil Company (20 per cent).

The project financing was done entirely via the balance sheets of the joint venture partners. Construction costs of the plant also included complete gas facilities from its own oilfields and a new 330 kilovolt (kV) transmission line to evacuate power from the plant to the grid. Fuel risk was borne 100 per cent by the joint venture (JV) partners with the fuel cost embedded in the US$0.022/kWh energy charge. The energy charge and the capacity charge (at approximately US$13.00/kW-Month) are without adjustment other than for inflation using OECD CPI. The plant has also come under fire for its higher than expected final investment costs. Parties are presently seeking to resolve the dispute directly (i.e. out of court). Meanwhile, the plant, which came on stream in 2005, is producing power, but, due to the dispute, full payment is not being made by PHCN. Therefore, as originally agreed to in the PPA, the plant will not amortize after five years.[86]

Although recently hailed as "a beautiful bride", with the President's economic advisor urging that "today not tomorrow" [is] the time to invest in Nigeria," [87] the country has a muddied track record when it comes to IPPs (Games 2006). Still, an additional IPP is due online, in 2007. Negotiations started in 2001 with Shell Petroleum for the third IPP, which involves a brownfield and greenfield investment namely: refurbishment of the existing 270 MW (Afam V) under an acquire operate own contract and the addition of 630MW (Afam VI) under a BOO arrangement. The project has, however, been delayed due to the civil unrest in the Niger Delta. Furthermore, more IPPs may be in the pipeline, including possibly a coal-fired plant at Enugue or Benue State, a new hydro facilities at Mambilla or Zungeru as well as sale and expansion of the state-led Ibom project,[88] particularly as the state aims to treble its capacity by the end of 2010 (from 5,198 MW at present to 15, 853 MW). Furthermore, four Nigerian firms (Farm Electric, Supertek, ICS and Eithope) have all been licensed to build power plants as of August 2006, with the expectation that power will start coming on line

[86] Interesting to note that there have been several IPPs sponsored by oil companies throughout South East Asia. Evidence gleaned by PESD researchers (as referenced in footnote 5), indicates that often such companies see IPPs as a means of marketing their primary product, namely oil, and have been more flexible in terms of negotiations related to power investments.

[87] As discussed in section 3.5.3, in January 2006, Nigeria received its first ever rating from Fitch Ratings, a BB- (3 notches below investment grade), followed by a comparable rating from Standard & Poor's (*Nigeria launches power privatization sales pitch* 2006).

[88] Although widely discussed in the media as IPPs, neither the 150 MW Omoku gas-fired, single cycle plant, nor the 180 MW Ibom gas-fired open cycle plant, both brought online in 2007 are IPPs following the definition of this paper. The plants were not funded by PHCN, the state utility, but there was no private money per se as they were 'independently' financed by state governments (Rivers State Government and Akwa Ibom State Government). As of end-2006, however, Globeleq is "pursuing" investment in the Ibom project, which would include both investment in the existing capacity and an addition of up to 500 MW (Globeleq 2006:7).

from these firms within the next four to five years, funded in part through domestic sources. With a regulator established since 2005, it is unclear the extent to which new and existing deals will be affected.

In May 2005, the Nigerian Electricity Regulatory Commission (NERC) was established; ostensibly independent from government with an independent source of funding, NERC has decision-making powers on key aspects of the ESI, such as approval of capacity expansion plans and oversight of quality of service. NERC is also charged with issuing licenses to companies operating in the sector and expected to regulate wholesale and retail electricity tariffs and prices. NERC's interface with the IPPs is still being mapped and tested. In the meantime, a special purpose entity (SPE) has been created as an interim arrangement post-unbundling with the mandate to take over some of NEPA/PHCN's liabilities, including the AES, Okpai and Afam contracts. This is intended to help ensure that unwarranted liabilities are not passed on to the transmission company.

4. Understanding the experience of IPP investments in Africa

In sum, how may one understand this sample of IPPs described above? There are two natural fault lines. The first cuts between North Africa and Sub-Saharan Africa, and the second, between those projects in SSA that faced some form of contract change and those that did not.

The main difference between IPPs in North Africa and those in SSA is that the former took root in significantly more attractive investment environments than the latter. In turn, competition was greater, as were ICBs, which may have also contributed to cost reductions. Also of significance in most of the North African projects is that despite exogenous stresses, policy frameworks remained largely intact and planning mishaps were fewer than in the SSA sample. Most North African IPPs benefited from abundant low cost fuel and secure fuel contracts as well as credit enhancements such as sovereign guarantees. For the most part, with few exceptions, these contributing elements to success were absent in SSA. Certain SSA projects did in fact benefit from ICBs, as well as abundant fuel resources and credit enhancements, but generally speaking, the abovementioned elements were largely consistent across only the North African pool.

Although distinguished by relatively less attractive investment climates, the SSA IPP sample is not homogenous. What if any are the shared characteristics among the seven projects (Iberafrica, OrPower4, IPTL, Songas, Takoradi II, AES Barge and Okpai), eight if Westmont is included, that underwent some form of contract change? Furthermore, does the balance of SSA IPPs that did not see any form of change (Tsavo, Mtwara, CIPREL and Azito) exhibit any common elements?

The IPPs that have seen changes to their original contracts were all procured amidst a form of electricity crisis, mostly as a result of drought in largely hydro-dependent systems. Furthermore, often World Bank sanctioned power master plans and/or conditional loans were passed over to plug the immediate power crisis. In such crisis situations, ICBs were only followed in two of the eight IPPs; furthermore, in the two projects that did see an ICB (Songas and OrPower4) only two bids were received. Such crisis conditions did not predominate to the same degree in the group of IPPs that have not seen changes. Although ICBs were still limited (Azito and Tsavo), it is arguable that the IPP framework was more clearly defined, bidding was more transparent and ultimately more thorough due to the fact that plants were procured under less urgent circumstances than in the group of eight discussed above.

Furthermore, there is a discrepancy in the prominence of DFIs and/or development-minded firms in the two different groups of SSA IPPs evaluated by this paper. In all four of the projects that have stayed intact to date in SSA, multilateral and bilateral development institutions and/or development-minded firms have been equity holders (IPS and IFC in Tsavo, the Netherlands Development Company (FMO) in Mtwara, IFC in CIPREL, and Globeleq and IPS in Azito). In addition, most of these projects have also benefited from concessionary funding, provided and/or arranged by DFIs. In contrast, in the eight remaining projects, which have encountered some form of change, such agencies and firms have been notably less present on both the equity and debt side. Only one of the eight projects had a development minded firm as an equity holder (Globeleq in Songas).[89] Only one of the eight projects saw involvement of a DFI on the debt side (World Bank in Songas).[90] Finally, it is important to here that Songas' contract changes were linked to the delays caused by the negotiation and ultimately arbitration of IPTL.

The two elements described above, namely the electricity crisis and the prominence of DFIs and/or development-minded firms are (like the favourable investment climate of North Africa) by far the most far-reaching elements in terms of differentiating the two groups of SSA IPPs evaluated here. There are, however, a couple additional elements that help explain some of the variation in the two groups. The absence of abundant low cost fuel also characterizes five of the eight projects that saw changes (Westmont, Iberafrica, IPTL, Takoradi and AES Barge), with only one project in the group of four (Tsavo) that have not seen changes not benefiting from abundant low cost fuel. Also worth mentioning is the fact that four of the eight ended up funding projects entirely with their balance sheets (Westmont,

[89] However, Globeleq only came into Songas, after the Allowance for funds used during construction (AFUDC) had accumulated to over US$100 million and is credited in part with allowing the buying down of the AFUDC by the government. For a detailed discussion of the AFUDC see Gratwick, Ghanadan and Eberhard (2006).

[90] Furthermore, OrPower4 did obtain a MIGA guarantee.

Iberafrica, OrPower4, and Okpai). Interestingly, although PPAs, credit enhancements, security arrangements, positive technical performance and strategic management have all emerged to shape particular projects, there is no fully conclusive data across the pool. Also noteworthy is the fact that of the four projects that have seen no changes, there were no local partners, including no participation on behalf of the government. In contrast, of the eight that did see a change, local partners, including government were present in six of the projects.

What then is the relationship between these elements and the sustainability of projects and contracts? Does a greater balance between development and investment outcomes enhance sustainability, and, if so, what are the contributing elements to success? The North African pool of projects is cited by stakeholders as having relatively favourable outcomes for both sponsors and host countries alike. Furthermore, the relative absence of contract changes confirms this conclusion. In contrast, SSA projects have been repeatedly cited as being out of balance by myriad stakeholders, and here again the litany of contract changes would seem to confirm this. Many elements have contributed to the North African successes, especially favourable investment climates, the endurance of policy and planning frameworks, even in the face of exogenous stresses, international competitive bidding procedures and low cost fuel. Furthermore, there are several key elements that differ between the SSA pool of projects that have faced contract changes and those that have not, most notably the emergency environment and the presence of multilateral and bilateral development institutions and/or development-minded firms. Not only have the above noted elements been instrumental in creating a greater balance at the start of the IPPs, they appear to have enhanced the sustainability of projects and contracts over time, particularly in the face of exogenous stresses.

4.1 Building up the contributing elements to success, at the country level

4.1.1 Favourable investment climate

The investment climate is not a deal breaker per se. IPP investments have been made in challenging contexts, most notably Kenya, which faced an aid embargo during its first wave of IPPs. However, the investment climate determines the degree of interest by foreign investors and hence the competitiveness of bids and ultimately the cost of the projects. The most favourable investment climates have been characterized as those with macro-economic stability, an active capital market, an efficient banking system, a history for upholding contracts and where recourse to arbitration is easily available. The relative absence of corruption, the availability of a well educated and productive labour force at reasonable rates and a growing economy with a focus on increasing the private sector's role also contribute. Finally, the financial health of the power sector, namely the solvency of the off-taker, together with the prospect of more than one investment may mean that less costly risk mitigation techniques may be required and that the cost of capital is reduced and reflected in the target return on equity.

Egypt, Morocco and Tunisia distinguished themselves from their Sub-Saharan Africa counterparts, with Tunisia scoring the highest in terms of its investment track record as evidenced by its investment grade ratings for foreign currency (BBB) and local currency (A). Both Egypt and Morocco score just one notch below investment grade (at BB+).

In contrast, the Sub-Saharan IPP sample has no investment grade ratings. Of the three countries that have received a speculative grade rating (Ghana, Kenya and Nigeria), two of these ratings (Kenya and Nigeria) were given in the last two years, long after IPP deals were signed, with Kenya's investment climate defined by its aid embargo in the mid-1990s, as previously noted. Furthermore, Ghana's speculative rating (at B+) is four notches below investment grade and therefore not comparable to the speculative grade ratings of Egypt and Morocco. Tanzania is also worth mentioning in this context. Throughout the 1990s, all export credit agencies were off-cover in Tanzania; no foreign commercial banks were willing to lend, as there was no clean track record of commercial loan repayment. Consequently, the possibility for a traditional project-financed IPP deal in this climate was limited. To summarize, for each of the three North African IPP success stories described here, countries have either had an investment grade rating or one notch below, in contrast to the five Sub-Saharan Africa IPP cases, where no country has received an investment grade rating and even speculative grade ratings have been few and far between.

Furthermore, despite the difference in the perception of the investment climate between North and Sub-Saharan Africa, incentives offered to investors in IPPs were relatively similar across the pool of eight countries and 20 projects, with some variety with regard to tax breaks.[91] For instance, nearly all 20 projects appear to have benefited from both customs and VAT exemptions during construction as well as full repatriation of profits. Currency conversion was also provided for in virtually all of the projects. In terms of tax holidays, all three countries in the North African sample had tax holidays of five years. In East Africa, Tanzania provided a tax holiday of five years, but Kenya's tax holidays only extended until plant commissioning. Although one would expect the investment incentives to increase with the perceived risk (with increased incentives offered in Sub-Saharan Africa), such a pattern is not evidenced.

How did the perception of the investment climate impact on project development? Quite simply, with demand for IPPs outweighing supply, those countries with a better investment

[91] Although many investment incentives were similar, there was considerable variation in terms of the security arrangements and credit enhancements that firms negotiated, including within specific countries as will be highlighted in section 4.2.4.

profile (primarily the North African sample) attracted more investors and ultimately were able to cement deals on more favourable terms to the host country. While not the only factor in influencing outcomes, the investment climate context goes a long way in setting the stage for negotiations and more balanced contract terms and helps explain the initial imbalance in so many of the Sub-Saharan cases.

4.1.2 New policy frameworks and regulation

An ideal framework involves a clear policy statement backed by legislation that: elaborates where and how power sector developments fit into overall energy policy; clarifies the governance of state-owned utilities (including corporatization and commercialization, transparent ring fencing of costs and adequate cost-recovery, together with a plan for possible unbundling); and makes room for private sector participation, including IPPs. Furthermore, the framework with regard to IPPs, ideally: defines the relative role of IPPs and the incumbent state-owned enterprise; specifies how IPPs will be procured and which agency has responsibility; introduces international competitive bidding standards; establishes an independent regulator and defines the powers and functions, among them the licensing of IPPs and approval of PPAs, including pass-through costs, and dispatch rules.

Although all eight countries in the sample have introduced legislation to allow for private generation, few countries have actually formalized and then realized a clear and coherent policy framework. Thus there is abundant evidence of tentative experimentation with private power that does not always lead to a sustained opening of the electricity market for private investment. Furthermore, long-term PPAs have the potential to constrain wholesale competition in the future, although means to transition to wholesale competition with IPPs have also been identified (Woolf and Halpern 2001). In addition, state-owned utilities are rarely exposed to market costs of capital and direct comparisons of their costs with IPPs are often difficult to discern.

Nowhere is the standard reform model for power sector reform being adopted full-cloth, namely unbundling of generation, transmission and distribution, introducing competition and private sector participation at both the generation and distribution level (United Nations Environment Programme and United Nations Economic Commission for Africa 2006:67; Malgas, Gratwick et al. 2007a; Malgas, Gratwick et al. 2007b). All utilities evaluated were 100 per cent state-owned with the exception of the Kenyan utility, which is owned 48 per cent by the state, and the Nigerian utility, which is in the process of being privatized (however, this development occurred after the two contracts evaluated in this paper were bid, and PHCN still maintains a coordinating role).[92] Thus, state-owned utilities remain in a dominant place, with IPPs on the margin and the future frameworks in many countries as of yet undecided.

The most coherent policy framework exists in North Africa, with Egypt defining itself as the frontrunner. In Egypt, 15 IPPs were specified by the EEA (later the EEHC), which was clearly charged with carrying out the IPP programme following ICB practices. IPPs were to provide the majority of new generation capacity to the grid. They were also to assist in benchmarking state-owned and operated plants and improve the overall performance of the sector. A regulator was to be integrated into the ESI and assist in oversight. Corporatization and commercialization of distribution assets was to follow. Of the numerous features of the policy framework, it was perhaps the clear mandate given to the EEA/EEHC to procure IPPs that helped define its early success. Even with a clear policy and the power to implement decisions, however, Egypt's IPP programme was derailed (after the first three plants were bid out), due to macroeconomic shock and a severe currency devaluation. In shelving the remaining plants, the government sought to reduce additional foreign exchange risk. Furthermore, reform of the incumbent state-owned utility has been slow. Thus, policy was either not implemented or was rewritten during implementation, a practice evidenced across the sample, with the present policy framework in Egypt favouring a return to publicly-financed generation.

While more piecemeal than Egypt, Kenya's policy framework has yielded one element that the other seven countries in the sample have presently parked or only pursued late in the game, namely the establishment of an independent regulator to aid in sector oversight. In Kenya, the regulator, together with the adoption of ICB practices, has helped to radically reduce PPA charges (between the first set of IPPs negotiated and the second). Kenya's ERB has also been instrumental in helping to set tariffs and manage the overall interface between private and public sectors. In Cote d'Ivoire, Egypt, Ghana, Nigeria and Tanzania regulatory agencies have come into force only after IPPs have been negotiated, and there has been little impact as of yet, with, in the case of Egypt, the regulator's powers severely limited. Finally, no attempt to establish a regulator has been made in either Tunisia or Morocco, with the ministries the de facto regulators in these two countries. What has emerged as a general trend is that the mere presence of a regulator is not in and of itself a defining factor. An independent regulator may have positive, negative or no impact on outcomes. If, however, regulatory governance is transparent, fair and accountable, and if regulatory decisions are credible and predictable, then there is greater potential for positive outcomes for host country

[92] In addition, it should be reiterated in this context that Cote d'Ivoire has had its state utility under a concession contract since 1990—an arrangement that is expected to continue until 2020. Tanzania has just completed a management contract of four years, and Kenya has just begun one.

and investor alike. Evidence also points to the fact that effective regulatory oversight may lead to a reduction in the stated capital costs of projects for selectively bid projects (Phadke 2007:10,25).[93]

A final policy and practice is worth noting in this context: in three of the eight sample countries (Nigeria, Tanzania and Tunisia) efforts have been made to exploit stranded gas as part of the IPP programme.[94] [95] In Nigeria, a reduction in gas flaring is central to the push for gas-fired power. In Tanzania, the IPP programme commercialized previously stranded (although not flared) gas via Songas and Mtwara. In Tunisia, although the primary goal was to attract new investment into the hydrocarbon sector, one significant spin-off has been the reduction of gas flaring. Each country has seen a distinct set of challenges,[96] however, generally this larger policy has insulated projects from intense public scrutiny with project sponsors and policy makers alike able to point to the benefits of the commercialized gas and the reduction in fuel imports.

Behind many of these policies sit the development finance institutions, notably the World Bank, which has had a hand in nearly all power sector reform programmes in Africa. These institutions were particularly instrumental in advancing private sector participation in generation. As many of those same institutions began reconsidering publicly-funded infrastructure investments again at the end of the decade, countries have often followed with policies that reflect this movement—from state to market and back again.

For many of the sample countries, however, the future framework remains uncertain, even as there appear to be more concessionary loans available. As one policy maker indicated: "The government is reviewing the impact of IPPs and the trend in countries like [ours]. [The utility] needs a breathing space. Soon[er] or later, the government will publish a strategic paper on the way forward." In the meantime, in this same country, emergency power has been ordered to plug an immediate power crisis, as will be further explored below.

[93] Furthermore, alternatives to strictly independent regulation are increasingly being considered, which may provide a better match to a country's regulatory commitment and institutional and human resource capacity. Alternatives include: regulatory contracts, the outsourcing of regulatory functions, expert panels and regional regulators (Eberhard 2007:14).

[94] Domestic gas reserves were used for IPPs in both Cote d'Ivoire and Egypt, however, unlike for the other countries mentioned above, this did not represent the establishment of a new gas infrastructure.

[95] An attempt was also made to exploit stranded gas reserves in Ghana's Osagyefo Barge project, which, however, has been led by the state, with as of yet no private participation, and no power produced

[96] In Tunisia, gas quality has proven poor and water also flooded the wells; consequently the plant has been repeatedly offline; in Nigeria, although the gas is of good quality, stakeholders have seen costs escalate which in turn has caused the utility to withhold payments. In Tanzania, costs have also escalated, but for reasons unrelated to the gas project itself.

4.1.3 Coherent power sector planning and execution

Intricately connected to sound policy frameworks are coherent power sector plans. Ideally, the latter follows from the former and includes four components: setting a reliability standard for energy security; completion of detailed supply and demand forecasts; a least cost plan with alternative scenarios; and clarifying how new generation production will be split between the private and public sector as well as clarifying the requisite bidding and procurement processes for new builds. Among the most important aspects of coherent power sector planning is vesting planning and procurement in one empowered agency to ensure that implementation takes place with minimal mishaps.

The sample evaluated in this paper has several noteworthy planning mishaps. In evidence are examples of demand and supply not being accurately forecast due partly to extended droughts, which in turn necessitated fast tracking IPPs, i.e. plants were sped through to meet immediate power shortages. The first two plants in Kenya (Westmont and Iberafrica), the first plant in Nigeria (AES Barge) and Ghana's IPP were negotiated amidst drought conditions. Generally, the speed was at a cost. Although both Westmont and Iberafrica came online within eleven months, later they were the source of scrutiny and investigation (due to un-transparent bidding practices and what were perceived as unnecessarily expensive charges). Furthermore, Westmont did not secure a second PPA due to disagreement over a tariff, with public stakeholders unwilling to make similar concessions a second time. In the case of Nigeria, although fast-tracked, the AES Barge took nearly two years to come on line due to a renegotiation of the PPA.

An inability to accurately estimate demand and supply as well as set a clear reliability standard has also necessitated several cases of emergency power where units have been ordered for one to two years with the purpose of plugging a short-term power crisis. In both of the East African countries in this sample as well as Ghana, the governments have ordered units to address drought and system collapse. In Kenya, the country harnessed 100 MW of emergency power twice: in 1999-2001 and again in 2006 (supplied by Aggreko, Cummins and Deutz in the first instance and Aggreko alone in the second). Emergency power has been turned to repeatedly in Tanzania also. In Ghana, emergency power was instrumental in reducing the impact of the 1998 drought, but with drought conditions reversing, the state failed to honour its contracts with SIIF Accra, which as of 2007, seven years later, remains an unresolved conflict. Costs for this emergency power, at approximately 30-40 US cents/kWh, are high, however, they are still less than the cost of no power (IFC per com 2005). As previously noted, Tanzania has estimated that it has saved around US$1.00 for every kWh of outage averted (or about five to 10 times the ordinary cost of generating electricity).[97]

In Tanzania, the speeding through of one plant has resulted in perhaps the most high profile IPP story on the continent to date. In this project, absent are critical planning elements, namely the lack of: a clear reliability standard, an accurate demand and supply forecast, a detailed plan for privately powered and publicly powered generation and most importantly a defined agency to implement the plan.

The Songo Songo gas-to-electricity project was in the Power System Master Plan, initially slated to come online within six months. However, the project was slow to materialize given its technical and financial complexity. With deadlines passing and power cuts persisting, it is alleged that other ministries, affected by the power cuts, started second guessing that TANESCO and the Ministry of Energy and Minerals, following the World Bank procurement procedures and relying on concessionary loans, would be able to deliver the project on time to address the shortages. As noted previously, the cost of unserved electricity to the economy was high and therefore Tanzania paid dearly for no power. Thus, the backdrop of the IPTL agreement appears to have been a failure to deliver on the Master Plan and hefty associated costs for many Tanzanians facing loss of services, TANESCO facing loss in revenue, and the Tanzanian economy facing loss of productivity, together with a clear interest in collaborating with Malaysian investors in the context of South-South partnerships.

The impact of this planning mishap was multi-fold: IPTL, which was negotiated quickly, behind closed doors, announced its total investment costs as US$150 million (US$163 including fuel conversion), which the Government of Tanzania and the World Bank would later argue was inflated by 40 per cent. This argument would in turn lead to a lengthy arbitration process spanning three years. During the time that IPTL was being disputed, the Songo Songo gas-to-electricity project would be put on hold, mainly through pressure from the World Bank, its largest donor, due to alleged corruption in the sector. Although the arbitration would ultimately lead to IPTL's investment costs being reduced to US$127 million, the cost was still above and beyond the price that the government sought to pay. Furthermore, due to the delays, Songo Songo accumulated US$100 million in interest charges on owner's equity, i.e. which the sponsor was owed by TANESCO. Additional costs to the state include emergency power that was required due to both IPPs being unavailable until 2002 and 2004, respectively.

Although it is easy in hindsight to accuse stakeholders of acting imprudently in the face of emergencies, the actual conditions of load shedding and shortages appear to have provided

[97] In terms of international norms, however, it should be noted that Tanzania's cost of unserved energy (CUE) is low. South Africa's CUE is approximately US$10, which is in line with the CUE in many industrialized countries (Global Energy Decisions 2007).

few alternatives. The solution appears to be in: taking steps to improve the investment climate, drawing up and implementing clear policy frameworks, (namely spelling out where and how private power fits into a single buyer model), building contingencies into the planning process, vesting planning in one agency, and conducting open bidding but under less cumbersome bidding procedures—all much easier said than done, but not infeasible for host countries to adopt and thereby move one step closer to balancing development and investment outcomes.

4.1.4 Competitive bidding practices

While policy and planning frameworks go a long way in determining outcomes, the type of bidding has been linked to outcomes, with considerable attention paid to the importance of international competitive bidding practices over selective bidding practices. Two recent studies, have evaluated the relationship, demonstrating that while there is evidence for ICBs leading to up to a 60 per cent reduction in the stated capital cost of plants, there is also evidence for selective bidding proving effective in certain instances, provided there is regulatory scrutiny (Deloitte Touche Tohmatsu Emerging Markets Ltd. and Advanced Engineering Associates International 2003; Phadke 2007).

ICBs were conducted for 11 of the sample of 20 IPPs, including all three projects in Egypt and Morocco, and one of the two projects in Tunisia. In the East African group, ICBs have been less common, with three of the seven projects (OrPower4, Tsavo and Songas) recorded as having conformed with such bidding practices. In West Africa, an ICB was conducted for only one of the five projects (Azito).

In terms of gleaning meaning from ICBs versus selective bidding, of the six projects that have faced renegotiation, four of them were bid selectively rather than via an ICB (IPTL, Iberafrica, AES Barge and Okpai), with the two exceptions being Songas and OrPower4. The absence of regulatory scrutiny is noteworthy in each of these four projects as well. Furthermore, Westmont, which was selectively bid, quit the country after its first seven year PPA expired. The other selectively bid projects have also, with the exception of CIPREL and Mtwara, encountered some difficulty or another, which has led to a change in how the project is being developed.[98] Ghana's Takoradi II has not been able to raise the finance for the second phase of the plant, and Tunisia's SEEB has not been able to secure its fuel supply and is presently offline. Although reasons for these stumbling blocks may be traced well beyond the presence or absence of an ICB, the correlation is nonetheless revealing.

[98] Although Tullow Oil, which initially won the contract to develop the Mnazi gas field in 1994, did encounter difficulty, this related primarily to the upstream portion of what is today known as the Mtwara Energy Project. Artumas Tanzania (Jersey) Limited (ATJL) has indicated that since negotiations began in 2003 it has encountered no such difficulties.

Furthermore, it should be noted that the success of the ICB process is intricately linked to the number of bids received, with more bidding driving down prices. The number of bids submitted to ICBs in North Africa was generally double to triple those submitted to ICBs in East Africa--with only three bidders in Kenya's Tsavo plant and two in both OrPower4 and Songas plants. All three projects have since been pressured to lower tariffs, as discussed repeatedly. In addition, the time and associated cost required to complete an international competitive bid should not be underestimated, with drought related energy crises often cited as the reason why ICBs have been passed over. Just as alternatives are being considered for strictly independent regulation, including contracting out, to match the institutional and human resource capacity in a country, the Sub-Saharan African examples here point to the need for more efficient bidding processes that while focussing on transparency and oversight also expedite timely outcomes.

4.1.5 Abundant low cost fuel and secure fuel contracts

The availability of competitively priced fuel supplies for IPPs has also emerged as a key factor in how IPPs are perceived and ultimately whether there is public appetite for more such projects, in large part because fuel is generally a pass-through cost to the utility and in many cases to the final consumer as well. Thus, if the IPP uses a fuel different from the incumbent fuel, and if that fuel is more expensive, then there is greater potential for stress on the project.

In three of the sample countries (Ghana, Kenya and Tanzania,), at the inception of IPPs, low priced hydropower was the dominant fuel source. In these countries, IPPs were thermal powered, using a combination of imported fuel oil and domestically procured natural gas. IPPs helped the countries to achieve greater fuel diversification, however, when the costs of IPPs (other than those running on domestically procured natural gas, namely Songas) were compared with state-owned, generally amortized hydropower, the new privately owned generation was seen to be largely more expensive, due partly to the energy/fuel charge. Furthermore, these countries witnessed a series of debilitating droughts over the course of the 1990s. Drought has also wreaked havoc throughout the East African region between 2002 and 2006. During this time, the existing hydropower infrastructure proved insufficient, and thermal, provided almost entirely by IPPs, was increasingly integrated into the fuel supply mix (from 10 per cent to 60 per cent in Tanzania), forcing up the price of power. Although more thermal power may be required, the public perception is that IPPs drive prices up, rather than a number of factors, including drought, which means that gaining public support for such projects is all the more challenging.

Contrast this story with Morocco and Nigeria. In Morocco, at the inception of IPPs, oil and coal, largely imported, were the incumbent fuels. Hydropower also played an important role in contributing to the generation mix. Through Jorf Lasfar, the country's first and largest IPP, Morocco changed its fuel composition; as of 2005, coal had become the dominant fuel, accounting for more than 60 per cent of generation. Although only 50MW, the country's wind power plant also contributed to the diversification, together with Tahaddart, which introduced over 300MW of natural gas-fired generation. Thus, like Kenya and Tanzania, Morocco has achieved fuel diversification through its IPPs, however, unlike the two other countries, Morocco has seen prices come down, due to a host of factors, including the relatively cheaper price of imported coal, and the use of gas via the Algerian-Spain pipeline, which fuels Tahaddart and which the government receives as a royalty.[99] Nigeria has relied entirely on domestically procured natural gas, and gas is the incumbent fuel. Until recently, although a series of issues affected project outcomes, most notably the investment climate and bidding procedures, fuel had been of no issue; however, recent civil unrest in the Niger Delta has led to a disruption in the fuel supply.

At the beginning of this section, the claim was made that when IPPs use fuel that is either cheaper than the incumbent fuel and/or the same as the incumbent fuel, IPPs have a greater chance of success. There are, however, several noteworthy exceptions. Tunisia's SEEB plant is one such exception. Although natural gas is the incumbent fuel in Tunisia, the SEEB plant was part of an initiative to attract investors to exploit stranded gas associated with oil production. Thus, since SEEB's fuel supply has been compromised, there has been no other source of supply for the plant. In Tanzania, the natural gas from the Songo Songo field, which was dedicated to supply the Songas plant and later to fuel IPTL, is cheaper than the imported fuel oil presently powering IPTL. There have, however, been significant delays in fuel conversion of the IPTL diesel units. Finally, in the case of Egypt, natural gas is the incumbent fuel and also the fuel of choice for all IPPs to date. Fuel is a pass through to the utility, but presently not to the final consumer. Fuel is also subsidized. In the last few years, the country has emerged as the sixth largest liquefied natural gas (LNG) exporter in the world. Presently a debate is raging about how to allocate the remaining natural gas reserves. Should they go for additional electric power generation, LNG export, or present and future industrialization projects? The issue then is not simply whether a country has abundant, low cost fuel, but whether security of supply is guaranteed through fuel contracts well into the future (for up to 30 years in the case of Jorf Lasfar and on average 20 years for the other projects). Fuel must be abundant and low cost, both now and later, for it to have a positive, not negative, impact on outcomes.

[99] Other contributing factors to a reduction in tariffs in Morocco include: economies of scale that Jorf Lasfar offers (in 2004, 66% of the country production was from one single plant), the reduction in government fuel taxes (undertaken at the start of IPPs to make tariffs more competitive with region) and the increased management efficiencies at ONE (technical as well as economic).

These many country level factors are summarized in the table immediately following.

Table 2: Contributing elements to successful IPP investments within the purview of host governments

CES	Details
Favourable investment climate	-Stable macro-economic policies -Legal system allows contracts to be enforced, laws to be upheld, arbitration -Good repayment record and investment grade rating -Requires less (costly) risk mitigation techniques to be employed which translate into lower cost of capital and hence lower project costs and more competitive prices - Potentially more than one investment opportunity
Clear policy Framework	-Policy framework enshrined in legislation -Framework clearly specifies market structure and roles and terms for private and public sector investments (generally for single buyer model, not, yet, wholesale competition in African context) -Reform-minded 'champions', concerned with long-run, lead and implement framework
Clear, consistent and fair regulatory oversight	-Oversight improves general performance of private and public sector assets -Transparent and predictable licensing and tariff framework improves investor confidence -Cost-reflective tariffs ensure revenue sufficiency -Consumers protected
Coherent power sector planning	-Energy security standard in place; planning roles and functions clarified -Power planning vested with lead, appropriate (skilled, resourced and empowered) agency -Power sector planning takes into consideration the hybrid market (public and private stakeholders and their respective real costs of capital) and fairly allocates new build opportunities among stakeholders -Planning has built-in contingencies to avoid emergency power plants or blackouts
Competitive bidding practices	-Procurement process is transparent and competition ultimately drives down prices
Abundant low cost fuel & secure contracts	-Cost-competitive with other fuels -Contract safeguards affordable and reliable fuel supply for duration of contract

Note: Competent contracting capacity is also emerging as a critical piece of the equation, which may lead to more favourable outcomes. Such capacity may ultimately be best located in a single-buyer office. Fair dispatch, namely the equitable dispatching of state-owned and privately-owned plants has also emerged as a critical piece of the equation, but is beyond the scope of the present study.

4.2 Building up the contributing elements to success, at the project level

Who were the investors and what did they do to navigate the varying investment climates as well as the changing policy and planning frameworks, including fuel supply? Starting with an evaluation of equity arrangements, this section examines trends in investor behaviour, and how investors secured revenue to service debt and reward equity, particularly in the face of exogenous stresses.

4.2.1 Favourable equity arrangements

Did the presence of local equity shareholders make a difference in project outcomes? Were projects with such participation less likely to face pressure from host country governments to change their contract terms? How did a firm's prior experience with a country play out in terms of the making and breaking of deals? What about the presence of development-minded firms such as IPS and Globeleq as well as DFIs? Table 7.1 immediately below lists each of the projects, followed by the country origins of sponsors and their respective equity share, whether projects faced a change in contract terms and finally if there was turnover of the majority (Mj) equity partner.

Table 3: Equity participation in IPPs

Project (country initial)	Equity partners (country, % of equity held)	Change in contract terms	Mj[100] equity turnover (#)
Sidi Krir (E)	InterGen (USA, 60%) & Edison (USA, 40%) sold to Globeleq (UK, 100%) in 2004-2005, conditional sale to PEL (JV between Malaysia Tanjong & Saudi Arabia Al Jomaih, 100%) in 2007	N	2
Port Said (E)	EDF (France, 100%) sold to Kuasa Power, subsidiary of Tanjong (Malaysia, 100%) in 2006	N	1
Suez (E)	EDF (France, 100%) sold to Kuasa Power (Malaysia, 100%) in 2006	N	1
Jorf Lasfar (M)	CMS (USA, 50%) and ABB (Swiss, 50%) sold shares to TAQA (UAE, 100%)in 2007	N	1
CED (M)	EDF (France, 49%), Paribas (France, 35.5%) & GERMA (France, 15.5%) maintained equity since 1997	N	0
Tahaddart (M)	ONE (Morocco, 48%), Endesa (Spain, 32%) and Siemens (Germany, 20%) maintained equity since 1999	N	0
Rades II (Tu)	PSEG (USA, 60%) sold shares to BTU Power Company (GCC[101], 60%) in 2004, Marubeni (Japan, 40%) retained shares since 1999	N	1
SEEB (Tu)	Centurion (USA, 50%) sold to Candex (Canada, 50%) in 2005, Caterpillar (USA, 50%) maintained equity since 2002	Y	1
Westmont (K)	Westmont (Malaysia, 100%) has sought to sell plant since 2004	-	-
Iberafrica (K)	Union Fenosa (Spain, 80%), KPLC Pension Fund (Kenya, 20%) since 1997	Y	0
OrPower4 (K)	Ormat (USA/Israel, 100%) since 1998	Y	0
Tsavo (K)	Cinergy (USA) & IPS (Int'l) jointly owned 49. 9%, Cinergy sold to Duke Energy (USA) in 2005, CDC/Globeleq (UK, 30%), Wartsila (Finland, 15%), IFC (Int'l, 5%) retain remaining shares since 2000	N	1
IPTL (Ta)	Mechmar (Malaysia, 70%), VIP (Tanzania, 30% **in kind**), both have sought to sell shares	Y	-
Songas (Ta)	TransCanada sold majority shares to AES (USA) in 1999 and AES sold majority shares to Globeleq (UK) in 2003[102]	Y	2

[100] Column indicates the number of times majority (Mj) equity share(s) traded hands, not minority. For instance in the case of Songas, majority shares have traded hands twice with three different lead sponsors, however, minority shares have been bought and sold many more times. See 71 below for details on Songas.
[101] BTU is owned by public and private investors in the Gulf Cooperation Council (GCC).

Mtwara (Ta)	Artumas (Canada, 80%), FMO (Netherlands, 20%)	N	0
CIPREL (C)	SAUR International, with 88% (JV between French SAUR Group owned by Bouygues, 65% and EDF, 35%) BOAD, (West African Bank for Development), PROPARCO, (the Investment and Promotions Company for Economic Cooperation, subsidiary of AFD), and IFC holding the remaining 12%; in 2005 all shares sold to Bouygues (France, 98%), except BOAD (2%)	N	1
Azito (C)	Cinergy (JV between Swiss ABB, 50% and French EDF, 50%) holds 65.7% of shares, CDC/Globeleq (11%), and IPS (23%)	N	0
Takoradi II (G)	CMS (USA, 90%), VRA (Ghana, 10%), CMS sold shares to TAQA (UAE, 90%) in 2007	Y	1
AES Barge (N)	Enron (USA, 100%) sold to AES (95%) and YFP (Nigeria, 5%) in 2000	Y	1
Okpai (N)	Nigerian National Petroleum Corporation (Nigeria, 60%), Nigerian Agip Oil Company (Italy, 20%), and Phillips Oil Company (USA, 20%) maintained equity since 2001	Y	0

Notes: N: no change in contract terms and/or in original project concept as laid down in PPA, Y: yes change in contract terms and/or original project concept.

Foreign firms were the dominant player in African IPPs. There were no exclusively locally sponsored projects, unlike in Malaysia and China where local IPPs abound (Woodhouse 2005:22-23,91). This should not be surprising given the limited capital available in countries across the sample, however it is worth noting, and it does raise the issue of foreign exchange exposure, treated in section 4.2.2 below.[103] Following this logic, there were only two projects in the pool in which local partners were the major stakeholder, Morocco's Tahaddart and Nigeria's Okpai. However, in both cases, the majority stakeholder was either the national utility or the national petroleum company. Furthermore, in the case of Morocco, ONE, the state utility initially intended to hold only a 20 per cent share and increased its share after EDF pulled out (Malgas, Gratwick et al. 2007a:16). In Okpai, the power project falls under the rubric of a state-led gas flaring reduction programme, in which international oil companies, presently operating in Nigeria, are being engaged in power projects. In the next such project, Afam, the Nigerian National Petroleum Corporation also has a majority share (of 55 per cent).

[102] Due to complexity, turnover is detailed in this footnote: Ocelot (Canada), TransCanada (Canada), TPDC (Tanzania), TANESCO (Tanzania), TDFL (Tanzania, sponsored by EIB), IFC (multilateral), DEG (German), CDC (UK) were shareholders by 1996, with TransCanada the majority shareholder; IFC and DEG sold shares to CDC in 1997/8; TransCanada sold shares to AES (USA) in 1999; Ocelot/PanOcean sold shares to AES in 2001; AES sold majority shares to Globeleq (UK) and FMO (Holland) in 2003. *After the AES sale, equity shares and associated financial commitments (expressed in US$ million) in Songas were as follows: Globeleq: US$33.8 (56%); FMO: US$14.6 (24%); TDFL: US$4 (7%); CDC: US$3.6 (6%); TPDC: US$3 (5%) and TANESCO: US$1 (2%). This does not reflect the additional US$50 million that Globeleq committed for the expansion.*

[103] Projects such as Osagyefo Barge in Ghana and several projects in Nigeria (Ibom and Omoku) have been financed by either national and/or state governments and have been loosely termed IPPs for the following reasons. In Ghana, private participation was anticipated, and in Nigeria, projects have been independent of the national utility, led entirely by the Rivers State Government and the Akwa Ibom State Government, respectively. As of end-2006, however, Globeleq is "pursuing" investment in the Ibom project, which would include both investment in the existing capacity and an addition of up to 500 MW (Globeleq 2006:7).

Local participation has been cited as a possible means to reduce risk (Hoskote 1995:11; Woodhouse 2005).[104] A total of seven of the 20 projects had local equity participation, namely: Tahaddart, Iberafrica, IPTL, Songas, Takoradi II, AES Barge and Okpai. To what extent then did such local participation impact favourably on outcomes? Of the seven projects, six have encountered some form of change to their contract. Furthermore, in four of these six projects, either the state utility or another government entity held an equity share, which would indicate that the mere existence of a local partner might not be critical in setting an original sustainable balance. What about in the renegotiating of terms, how might a local partner make a difference? Kenya's Westmont and Iberafrica were both negotiated at the same time under similar policy frameworks. They are the only two examples in the project pool where one had local participation (Iberafrica) and the other did not (Westmont). As has already been discussed, Iberafrica first voluntarily reduced its tariff and then went on to negotiate a second 15 year PPA, in contrast to Westmont, which quit after failing to come to agreement on a second PPA. The presence of a local partner may have helped in creating a longer term solution, however, with just one example, the evidence is not conclusive.

Origins, experience and mandate of partners

Although globally, IPP investments during the 1990s were led by a host of American and European investors who saw returns in their home markets diminishing, there was also a wave of investors originating from developing countries, particularly from Malaysia.[105] In both Kenya and Tanzania, this paper has profiled Malaysian firms committing to projects (including in one of the projects, Westmont, cited above where the firm took neither foreign nor local partners). In Egypt, EDF has recently sold its assets to Tanjong, a Malaysian firm, and Globeleq has entered into a conditional sale for its entire Asian and North African portfolio, which includes Egypt's Sidi Krir, to a joint venture between Tanjong and Saudi-based Al Jomaih.[106] [107] While it would be inaccurate to say that these firms overlooked the

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[104] Stanford PESD limits the benefits of local partners to "near-term and tactical business management, [the] study suggests that they play little role in grander tasks such as enforcing contracts," (Woodhouse 2005:98).

[105] Michael Schur, Stephan von Klaudy and Georgina Dellacha provide a preliminary analysis of the contribution of firms based in developing countries in infrastructure projects between 1998 and 2004, noting an overall increase and a particularly steep climb in transport investments by such firms. The authors conclude by indicating that the involvement of such firms is encouraging, however, the jury is still out on whether such firms are better positioned to deal with a host of complex political economy issues that characterize developing countries (Schur, Klaudy et al. 2006).

[106] Another example of an emerging economy firm taking market share is BTU, based in the Gulf Cooperating Council, which bought shares from PSEG Tunisia's Rades II plant, in 2004.

[107] There is evidence that Chinese and Indian firms may assume a greater piece of this market. India's Tata is among the bidders for South Africa's IPPs and Chinese firms are leading power projects in Sudan (Merowe dam project), and in Nigeria (Omotsho and Paplanto), (*China in Africa: from oil & gas to infrastructure* 2006).

higher risk profiles of the African continent (and/or did not ultimately charge higher returns), there may have been a greater willingness to consider investments in the first place.

While the number of developing/emerging country-based firms appears to be growing, three of the southern-based firms are trying to sell their shares (Mechmar and VIP in IPTL and Westmont). Thus, the origin of the firm does not mean that project equity is set for life, or that such a firm is best positioned to service debt and reward equity since each of these sales appears to be motivated in part by an inability to do just those things.

A more revealing aspect than firm origins appears to be a firm's experience and mandate. Across the pool, examples pile up of firms being actively involved in the country prior to their IPP investment. For instance, EDF had a long-term relationship with Egypt in terms of providing technical assistance. Union Fenosa, the parent company of Iberafrica, had an existing relationship with Kenya through an information technology (IT) contract. IPS, a major shareholder in Tsavo and Azito, had operated in Kenya since 1963 and in Cote d'Ivoire since 1965. CDC, from which Globeleq was spun off, had a 50 year history in Tanzania. While the long-term presence of a firm does not appear to be decisive (as many such projects did face contract changes), it may help explain why more contracts did not unravel. Long-term relationships, with strong local management appear to have contributed to the staying power of firms and often the rebalancing of contract terms.

The mandate of the firm also appears to play a central role in the investment decision as well as the terms of the deal. Until recently, the two firms that were increasing (rather than maintaining or reducing their stakes) were Globeleq and Industrial Promotion Services. Until May 2007, Globeleq held an 11 per cent share in Cote d'Ivoire's Azito, a 100 per cent equity stake in Egypt's Sidi Krir, 30 per cent equity in Kenya's Tsavo and 56 per cent in Tanzania's Songas. IPS holds a 23 per cent share in Azito, and together with Duke Energy, a 49.9 per cent share in Tsavo. IPS is also leading development of Uganda's Bujagali project, a 250 MW hydroelectric plant, which was formerly being developed by AES.

Although both Globeleq and IPS are driven by commercial interests, these firms have emerged from agencies with a strong commitment to social and economic development. Globeleq remains wholly owned by CDC, the private sector promotion arm of the Department for International Development (DFID). IPS is Aga Khan's Fund for Economic Development operating arm in the industrial sector throughout Asia and Africa. While projects for both firms must make commercial sense, they must also serve a clear developmental function for the country/community. It is this commitment to development that appears to be particularly helpful in the face of African risk.

None of the projects with involvement of such development-focused firms, save Tanzania's Songas, has seen any changes to contract terms, which may signal a greater perceived balance from project inception as well as a better ability to withstand public pressure.

Furthermore, in terms of the Songas change, although the buying down of the Allowance for funds used during construction (AFUDC) of US$103 million brought about a reduction in the capacity charge, the firm received full payment upon the buy-down and therefore it represents a different case than many of the contract changes cited above.

An important development must, however, be reiterated in this context. Globeleq, which until 2006 seemed to be expanding its portfolio of assets, is in the midst of selling off the brownfield plants it bought to in turn position itself as a global, greenfield developer. The firm always intended to take this course, as it saw its brownfield investments as part of a stop-gap measure during the global downturn in private power. Recent, indicative bids for Globeleq's Sub-Sahara African assets were not, however, deemed viable and therefore for the time being Globeleq is holding on to its stake in Azito, Tsavo, and Songas. On the one hand, Globeleq's sale of Sidi Krir and its Asian assets signals that there may indeed be a renewed interest in private power. On the other hand, the absence of favourable bids for Sub-Saharan Africa indicates that such renewed interest is still limited in scope (beyond supplying emergency power), and there may indeed be need for this type of development-minded firm in less developed countries.

Meanwhile the presence of DFIs persists in project equity. Although none of the North African projects saw such participation, four of the Sub-Saharan African IPPs saw DFIs pick up equity shares. IFC holds a 5 per cent share in Tsavo's equity. Until 2005, IFC also held, together with BOAD and the Investment and PROPARCO, a 12 per cent share in CIPREL. Both IFC and the German Investment & Development Corporation (DEG) had an approximately US$12 million equity investment in Songas, with both organizations selling their shares after the IPTL dispute became known. FMO maintains a 24 per cent share in Songas (excluding the expansion of 65 MW) as well as a 20 per cent share in Mtwara. CDC, independent of Globeleq, also holds a 6 per cent share in Songas (excluding the expansion). It should be reiterated here that none of these projects, save Songas, has seen any contract changes.

Equity turnover

Equity turnover represents a final piece of this evaluation related to equity arrangements and how such arrangements have impacted on outcomes. Of the 45 original equity partners in the sample pool, 15 have exited from 11 different projects. This statistic, however, tells only part of the story. First, as previously indicated, three shareholders have been actively trying to sell their assets (both shareholders in Tanzania's IPTL and Kenya's Westmont) for several years. In the case of IPTL, Mechmar, the lead shareholder, has indicated that the arbitration settlement ultimately hurt equity partners, which has motivated the sale. VIP, the minority shareholder in IPTL, cites the following causes: oppression by the majority shareholder;

fraud by Mechmar in inflating the IPTL capital cost; and failure by Mechmar to pay its equity contribution (i.e. the project was 100 per cent debt financed). There has been no resolution of this conflict, and no willing buyers. In terms of Westmont, the firm did not secure a second PPA, due to disagreement over tariffs and has, since 2004, been seeking to sell the asset. Second, if one focuses exclusively on majority shareholders, nine of the majority shareholders in the 20 projects have sold shares at least once and two of the majority shareholders have been actively seeking to do so for at least three years, as described above.

The repeated refrain from most sponsors is that the sale of assets is motivated primarily by changing circumstances in home markets and/or related to corporate strategy; that is, the sale has little to do with the host country actions and reactions and/or poor investment outcomes, namely the ability to service debt adequately and reward equity. InterGen's reason for selling its interest in Egypt's Sidi Krir to Globeleq in 2004 was based on the fact that its shareholders (Bechtel and Shell) made a strategic decision to move out of the business of owning and operating private power facilities. For Bechtel this meant moving back to its core business of designing, engineering and building plants, but not operating and maintaining them, and for Shell, it meant focusing on the petroleum exploration and production business. Similarly, Edison has sold much of its global portfolio because of a decision to return to its core business in Italy. EDF also cites its plans of concentrating its investments in Europe. In terms of Tunisia's Rades II plant, PSEG described Tunisia as "an excellent place to do business" and noted that "the sale in no way reflects any unhappiness with [our] experience in Tunisia, but rather is in keeping with the company's stated strategy of reducing its international risks by selectively selling assets if they can obtain an attractive price" (Malgas, Gratwick et al. 2007b:15).

How does this refrain square with the contract changes? The majority shareholders in two of the eight projects that saw contract changes exited after such a change (namely, CMS in Takoradi II and Enron in what is presently known as AES Barge). Furthermore, as noted above, Westmont has sought to sell the plant since it failed to renegotiate a second contract, and Mechmar has actively been seeking to sell its shares post-renegotiation. In addition, although no contract changes are in evidence, one should not overlook the larger IPP programme change in Egypt. All three original sponsors have sold their assets, following the failure of the utility to follow through with 12 additional IPP projects as originally specified, which could have given existing sponsors a larger market share.

While less than expected investment outcomes may be partially motivating sales, turnover does not in and of itself appear to be challenging the long-term sustainability of contracts since in nearly all cases sellers have found willing buyers to take over the original or recently renegotiated PPAs. The two exceptions here again are the Westmont plant, where the first

PPA has expired and which was shrouded in controversy and IPTL, which has been embroiled in lawsuits and therefore it may be understandable why the plants have not attracted buyers. One stakeholder went so far as to assert, “[equity turnover is a] healthy factor in a maturing market. It is a good sign when investors come and go – not a bad or threatening thing.” The return of the government as shareholder, as planned in the case of Tanzania’s IPTL as well as Globeleq’s decision to hold onto to its SSA portfolio, would, however, signal that some markets might actually be less mature than expected.

What in the end have been among the most critical characteristics of equity arrangements that have led to project sustainability? Overarching characteristics appear to be firms’ prior experience in a country, the presence of development-minded firms and development finance institutions.

4.2.2 Debt arrangements: global and local

With debt financing covering often more than 70 per cent of total project costs, low cost financing has emerged as a key factor in successful projects. How and where to get this low cost financing is the challenge, but possible approaches in the African cases lie in DFI involvement, credit enhancements, and some flexibility in terms and conditions that may allow for possible refinancing. The goal for sustainability appears to be that the risk premium demanded by financiers or capped by the off-taker matches the actual country and project risks and is not inflated.

While there is no uniform pattern in the debt financing for the projects considered in this paper, there are a series of trends for how investors handled costs as well as practices that may ultimately contribute to success. Important to note at the outset is that although non-recourse project financing is the norm for privately financed electric power plants in developing regions, this sample of twenty projects saw several notable exceptions, including Nigeria’s Okpai plant, which was 100 per cent financed by the balance sheet of equity partners, together with the second phase of Songas (however refinancing is presently being pursued in the latter case). Westmont, Iberafrica and OrPower4 were also all financed entirely with the balance sheets of their sponsors. For Westmont and Iberafrica, the reason cited for this arrangement was that insufficient time was available to arrange project finance as plants had to be brought online within 11 months. For Orpower4, reasons are linked, by the sponsor, to the lack of a security package, which was not forthcoming until 2006.

DFIs and their impact on projects

With limited appetite for projects among many commercial banks,[108] development finance institutions are conspicuous in providing debt to projects across the pool.[109] Such entities participated in nearly every IPP, including significant participation on the part of the World Bank/IDA (CIPREL, Songas), IFC (Azito, Port Said, Suez, Tsavo), CDC (Tsavo, Azito), EIB (CED, Songas), DEG (Tsavo, Azito), FMO (Azito), African Development Bank (Azito) and PROPARCO (CED). In addition a number of export credit agencies were involved in providing financing: the USA Export Import Bank (Jorf Lasfar), the Overseas Private Investment Corporation of the United States, OPIC (Jorf Lasfar), and Japan Bank for International Cooperation, JBIC (Rades II).

Much of this involvement is related to the long history of DFI activity throughout Africa coupled with the real and perceived risks across the continent, which preclude private investors from filling the financing gap. The involvement is also, however, linked to the broader mandate of power sector reform. Still, it is noteworthy that African IPPs, which by their very definition imply private investment, had such significant public involvement.

Although projects with DFI funding tended to take longer to reach financial closure, sponsors did cite clear benefits; multilateral and bilateral development institutions helped maintain contracts and resist renegotiation in the face of external challenges such as Egypt's currency devaluation and Kenya's droughts when developers were pressured to reduce tariffs. A particularly revealing contrast is in the two Kenyan plants, OrPower4 and Tsavo, negotiated under the same policy framework, including via ICBs. The former plant saw no multilateral involvement in either its equity or debt whereas for the latter, IFC arranged all the debt and took a 5 per cent equity stake. Tsavo has since resisted pressures to reduce its tariffs by KPLC and the government, with the presence of a multilateral development institution cited as among the reasons. OrPower4 on the other hand has ultimately reduced its tariff for the second phase of the plant. Tanzania's Songas project, for which the World Bank together with EIB provided all the project debt, also deserves special mention here. The project took almost a decade to reach financial closure, but the World Bank played an instrumental role, in among other things, pressuring the IPTL arbitration, which ultimately led to what has been widely perceived as more balanced contract terms.

Locally denominated finance: the cost and benefit

Locally denominated financing appears to be among the solutions for more sustainable foreign investment, however, capital markets in many African countries are insufficiently

[108] A recent review indicates that Africa has attracted less non-recourse bank debt relative to private investment in infrastructure than other developing regions (Sheppard, Klaudy et al. 2006).

[109] See footnote 1 for a definition of development finance institutions.

deep or liquid to provide such financing for all projects. As previously noted, only one project across the pool, Morocco's Tahaddart, a 384 MW CCGT, negotiated a locally denominated PPA due to the fact that all of its debt (€ 213 million) was sourced from local banks. This local financing was aided by a number of factors including: the state utility's prominent role in the plant (holding nearly 50 per cent of total equity) as well as the fact that Morocco's commercial banks have a significant degree of state involvement.[110] With or without state involvement, no other country in Africa, has, as of yet, managed to arrange this level and depth of financing for IPPs. Thus, Morocco stands as a pioneer in this respect.[111] [112]

The main drawback for IPPs without locally denominated finance may be seen as projects undergo the effects of macroeconomic shock and currency devaluation. In Egypt between the end of 2002 and early 2003, the currency lost half of its value and PPA payments in local currency terms doubled. In the short-term the US dollar and/or Euro denominated PPA represented among the greatest safeguards for sponsors, as equity and debt holders in Egypt did not see the value of their investments decline. The host country, however, paid dearly for the plants. Furthermore, one could argue that equity and debt holders did lose out indirectly due to the fact that a decision was made to cancel additional IPPs after the devaluation.[113] While no country other than Egypt in the sample experienced the crippling effects of macroeconomic shock, over the course of the decade, Ghana, Kenya and Tanzania saw serious creeping devaluation, with the currencies losing more than 100 per cent, 200 per cent and 400 per cent of their value, respectively, over the 1990s, which has inevitably had an impact on capacity charges.[114] There has been pressure to reduce such charges as well as to reconsider IPP development in each of these countries at different stages.

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[110] Banque Centrale Populaire (BCP) put up MAD1300m and MAD960m was extended by a consortium of banks consisting of BCP as the lead lender, the Banque Marocaine pour le Commerce Extériere (BMCE) and Crédit Agricole (CNCA). Average exchange rate for the Moroccan dirham in 2003, the year that construction started, was 10.95MAD=1.00EUR (Interbank Rate).

[111] Local currency financing has been recently employed by AES in Cameroon for the 80 MW Limbe power station. The local offices of Standard Chartered Bank, Ecobank, Afriland First Bank and the Commercial Bank of Cameroon provided short-term local currency financing for the AES project, which served as a bridge to longer term dollar denominated debt (The role of local banks in financing power projects 2005:1)

[112] Stakeholders involved in Namibia's Kudu gas-to-power project are also raising the possibility of local currency financing, which could slowly help to change the face of African power, by introducing a new level of financial sustainability for all those involved. Kudu involves commercializing domestic (Namibian) gas and the construction of an 800 MW plant, initially intended mainly for export to RSA.

[113] Egypt's Sidi Krir plant did obtain local currency financing, however, it was US dollar denominated and therefore this ultimately did not help assuage the effects of the currency devaluation for the host country.

[114] Morocco and Tunisia saw minimal fluctuation, and Nigeria's fluctuations were limited to the period before 1998, when the first IPP was negotiated. Cote d'Ivoire experienced a major currency devaluation in 1994, however, this predated the IPPs, and was prompted by a World Bank and the International Monetary Fund (IMF) proposal rather than an exogenous macroeconomic shock.

Although few projects have benefited from locally denominated financing, there are some promising signs, including in South Africa's IPPs. Sponsors for South Africa's two new peaking plants, totaling a combined 1000MW slated to come online in 2009, have been given access to the country's capital markets and capacity payments will be denominated in rand. Furthermore, as discussed in section 3.5.3, four Nigerian firms (Farm Electric, Supertek, ICS and Eithope) have all been licensed to build power plants as of August 2006 with the expectation that power will start coming on line from these firms within the next four to five years, funded in part through domestic sources.

Where US dollar/Euro denominated financing is the only possibility, the use of a foreign exchange liquidity facility may be one solution, as it requires the PPA to be indexed to the local currency inflation, rather than the foreign exchange rate, and hence power prices will not go up faster than domestic inflation.[115] In terms of the sample, however, the most wide-spread and effective practice witnessed to date is the indexation of payments to a basket of currencies, as seen in Morocco's Jorf Lasfar, CED and Tunisia's Rades II project, highlighted in section 3.3.3.

4.2.3 Securing revenue: The PPA and other security arrangements

All of the projects evaluated had a long-term power purchase agreement with the (majority) state-owned utility[116] to ensure a market for the power produced (with the exception of Mtwara).[117] Such a contract was demanded by equity and debt holders alike given the lack of liquid markets for electricity in the sample of countries. As a result, in most cases there was competition for the market, but no actual competition within the market once the PPAs were negotiated, with contracts averaging approximately twenty years.

In addition to indicating who would buy the power, the PPAs detailed how much power would be bought and at what cost. How plants would be dispatched, fuel metering, interconnection, insurance, *force majeure*, transfer, termination, change of law provisions, refinancing arrangements, dispute resolution were generally all clearly laid out as well. Nearly all of the contracts specified some form of international dispute resolution and all but one contract specified a minimum availability, with Tunisia's SEEB being the exception.

These contracts were in turn backed by a series of security arrangements, including in some cases escrow accounts, letters of credit, stand-by debt facilities, committed public budget and/or taxes/levies, targeted subsidies and indexation in contracts. For instance, the Tsavo

[115] There has, however, only been one such application to date, namely the AES Tiete project in Brazil. A standby credit facility of US$30 million was used to mitigate devaluation risk and closed in May 2001 (Matsukawa, Sheppard et al. 2003:19)

[116] Section 4.1.2 reviews the exceptions, namely Kenya and Nigeria.

[117] Mtwara currently operates under a two year interim PPA, which is expected will be replaced by a 20 year PPA, to be finalized in 2007.

plant in Kenya, in which IPS is a major shareholder, has an escrow account equivalent to one month's capacity charge and a stand-by letter of credit from KPLC, which covers three months billing of approximately US$12 million. It is known that a minimum of eight of the twenty projects had either an escrow account or a liquidity facility or both, with typical terms being between one and four months capacity charge in reserve (with one month most typical for North African countries and up to four months seen in Tanzania).[118]

Not surprisingly, the number of security arrangements and credit enhancements appears to diminish as risk profiles improve, however there are noticeable exceptions such as the first wave of plants in Kenya (Westmont and Iberafrica), where the risk appears to be entirely reflected in the (higher) capacity payments negotiated, however corruption was also alleged in both of these plants.[119] Thus, the 'security arrangement' may lie not in a formal escrow account, but in an informal agreement among sponsors.

In all but one case, sponsors negotiated or were granted outright US dollar or Euro denominated PPAs, thereby reducing project sponsors' exposure to currency devaluation, which in certain cases such as Egypt's devaluation was severe. This may in fact be considered among the greatest security for sponsors and one that ultimately has influenced countries to change their position about further IPP investments. The one exception is in Morocco's final IPP, Tahaddart, which as discussed above was an entirely domestically financed plant, and which therefore had a Moroccan dinar denominated payment stream.

How then did the PPAs and security arrangements fare over time? As discussed at the outset of this section, eight of the projects, nearly half, have faced some form of change to their contract. Thus, many have not proven iron clad. Although there does not appear to be a strong pattern between security arrangements and contract changes (meaning that projects with either more or less security arrangements have been more or less susceptible to change), there is evidence for contract changes being directly related to the terms of the PPA in six projects.

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118 Morocco's Jorf Lasfar has an escrow account equivalent to one month's capacity charge and Tahaddart has a letter of credit equivalent to one month's payment (which also serves as a form of a liquidity facility). Tunisia's Rades II has an escrow account, however, the amount is not publicly available. The security arrangement for Kenya's Tsavo plant is detailed above in the text, and OrPower4 has since been granted a similar security package. It was specified that Tanzania's IPTL would have an escrow account equivalent to between 2 to 4 months of capacity charge, but this account has not been established. Songas was granted an escrow account for the first 115 MW, with the Government of Tanzania to match every US$1 spent by the project company. No escrow account was required for the Songas expansion; furthermore the escrow account was used in part to help buy down the AFUDC. The project also negotiated a liquidity facility equivalent to four month's capacity charge for the first three years, declining to two months starting in year four through the remaining years of the contract. To support its interim PPA, Mtwara has an escrow account (amount unpublished), which will be replaced by a form of liquidity facility, termed the Tariff Equalization Facility, once the 20 year PPA takes effect. Cote d'Ivoire's Azito plant has an escrow account equivalent to one month's capacity charge.

119 In 2005 it was found that the then Managing Director of KPLC, Samuel Gichuru, received US$2 million from Westmont.

Costs in Kenya's first wave of IPPs were inflated in part due to the short duration of contracts (of only seven years). With Iberafrica facing ongoing pressure to reduce its tariff coupled with an interest in negotiating a second contract, the sponsor voluntarily reduced its capacity charge, enshrined in the PPA, which meant that it did not amortize the full cost of the project over the first contract. At 15 years, Iberafrica's subsequent PPA is notably longer than its first (and with negotiations presided over by the ERB, tariffs have been deemed significantly cheaper). The oft-referenced Westmont did not negotiate a second contract after it failed to obtain the same terms, namely capacity charges, spelled out in its first PPA. The changes in Kenya's OrPower4 and Tanzania's Songas have also been related in part to the final amount of the capacity charge (as originally spelled out in the PPA).[120] In terms of Nigeria's AES Barge, initially sponsored by Enron, the renegotiation of 1999-2000 brought about several changes to the PPA, including a change in the fuel specifications (from liquid fuel to natural gas), which led to a major reduction in the fuel charge for the off-taker. The present renegotiations with AES Barge involve among other things reconsideration of the availability deficiency payment. Under the existing PPA, liquidated damages recoverable for a shortfall in availability are only 30 per cent of the agreed unit cost and payable only at the end of the year. The current proposal tabled is to increase this per cent significantly as well as ensure that payments are made on a monthly basis. In each of the five cases reviewed here, it has been the original terms of the PPA that have in hindsight been viewed as unsustainable for the host country and hence challenged. The case of Tanzania's IPTL is slightly different. Although the contract was considered unsustainable due to the added capacity of Songas, the IPTL arbitration was prompted by what was deemed a breech in the PPA, namely the project sponsors' substitution of medium speed engines for slow speed engines, without passing on the capital cost-savings to the utility, as per the PPA.

Thus, the PPA has been a central document, however, not necessarily because it has been bombproof. Rather, it has been the focal point of many of the discussions when deals have been considered out of balance.

4.2.4 Credit enhancements and other measures

Of the many different credit enhancements and other risk management and mitigation measures, it is the provision for international arbitration and sovereign guarantees that have been most commonly employed. All projects in the pool specified some form of international arbitration. Sovereign guarantees were extended for seven of the 20 projects in the pool: all three of the Egyptian IPPs, Morocco's Jorf Lasfar, Tanzania's IPTL, Nigeria's AES Barge and Cote d'Ivoire's Azito and Ghana's Takoradi II. However, one of the projects without

[120] It is, however, worth reiterating in this context that failure to agree on both the security package and the capacity charge contributed to delays in the development of OrPower4's additional 36 MW.

guarantees (Rades II in Tunisia) was given added assurances by the government. Furthermore, in the case of the Okpai plant in Nigeria, security in the form of the state-owned oil company's revenues were extended. Thus, if the off-taker defaults, NNPC, among the most liquid firms in the country, is liable.

World Bank partial risk guarantees are seen in two of the projects: Jorf Lasfar and Azito. In the case of Jorf Lasfar, the partial risk guarantee protects the commercial lenders should the off-taker and the government fail to make specified termination payments. In the case of Azito, a partial risk guarantee (PRG) ensures that commercial lenders will be paid, even if the utility and state default on payments (of both interest and principal) (World Bank 1997; World Bank 1999). In both cases, it is only after there is a breech in the sovereign guarantee that the PRG is triggered (Sinclair 2007:36).[121]

In addition, other measures were engaged. AES Barge has political risk insurance provided by OPIC. OrPower4 has a MIGA guarantee. Jorf Lasfar has a political risk guarantees from the World Bank, the Italian Export Credit Agency and the Swiss Export Guarantee Agency. Despite the multitude of risks perceived, however, and the plenitude of risk insurances available, it is here that the list ends, with no cover for the majority of projects.

What then is the relationship between such credit enhancements and the sustainability of projects? To what extent have they been effective at attracting and/or assuaging lenders? And to what degree have such mechanisms helped keep projects intact or led to a swift resolution, in the face of external pressures?

For each of the Egyptian IPPs, sponsors have indicated that the sovereign guarantees were essential to the deal, given the novelty and size of the projects. There is also evidence for Azito's partial risk guarantee as being among the key's to attract commercial lending (World Bank 1999). The lack of sovereign guarantees has also been cited as the main obstacle for developing the second phase of Ghana's Takoradi II. In Kenya, the only country in the SSA pool not to extend any sovereign guarantees, stakeholders in Tsavo indicated that without such a guarantee, the presence of IFC became critical, both to help arrange debt and share in equity. Across the board, sponsors of the Kenyan projects a well as KPLC cite the absence of sovereign guarantees as hampering the ability to raise private finance. ERB's rejoinder to this charge is that IPPs have been introduced to help commercialize the sector; government guarantees work against this goal, and MIGA and other risk insurers are available to provide such cover.

[121] Among the more well documented World Bank partial risk guarantees has been one extended for Uganda's electricity distribution concession with Globeleq. Here, regulatory risk related to changes in exchange rates, namely the risk that the regulator will arbitrarily interfere in foreign exchange-tariff adjustments, has been mitigated through the use of partial risk guarantees (Globeleq 2006:21; Eberhard 2007:1).

Finally, in no projects have the sovereign guarantees, political risk insurance (PRI) or PRGs been invoked, including in those projects which ultimately have faced a change in the contract (namely AES Barge, IPTL, OrPower or Takoradi II). Recourse to international arbitration has only been made in the case of IPTL, with the arbitration serving to shave US$30 million off of the investment cost. In addition, there is evidence that a MIGA delegation was sent to ascertain facts in the case of Kenya when OrPower4 was pressured by both the government and KPLC to reduce its tariff, but the guarantee was never officially invoked. Although pressure from KPLC continued thereafter, pressure from the government subsided (after the MIGA visit).

4.2.5 Positive technical performance

Virtually all IPPs in the sample have been characterized by positive technical performance, with exceptions noted in the case of Tunisia's SEEB and Nigeria's plants due to fuel supply. IPPs performance is generally superior to state-owned plants, with the example of Kenya highlighted here as an illustration. In terms of availability, between 2004 and 2006, IPPs had an average availability of approximately 95 per cent versus KenGen's thermal plants, which averaged 60 per cent.

Positive technical performance has been instrumental in changing the way IPPs are perceived. Consider for example IPTL, which performed optimally throughout the recent drought and helped Tanzania to stave off load shedding until 2006. The plant has since been termed "a saviour", even by stakeholders who indicate that corruption was likely. At the same time, an unexpected break in Songas' power in 2006, amidst the drought, temporarily tarnished the projects reputation, despite the fact that the plant still reached its average availability as specified in the PPA and all costs were borne by Songas. During its ownership of Port Said and Suez, EDF noted that its management of technological risk, namely by not outsourcing to any firms and handling all contracts within the fully integrated EDF, was also a key factor in leading to relatively positive outcomes for the firm (EDF 2005).

4.2.6 Strategic management and relationship building

Once twenty year contracts are in place, it would seem that the deal is set and the revenue secured, with clear provisions to ensure debt repayment and reward equity. There are, however, several other interrelated actions that deserve mention here. One involves relationship building, including via local partners (as previously discussed) and community social policies adopted by sponsors. Yet another action relates to how sponsors handle the onset of stresses, including through capacity charges and refinancing.

In terms of social policies, numerous project sponsors have adopted outreach programmes to improve relations with local communities. For instance in Kenya, Tsavo power set up a

US$1 million community development fund for the duration of the 20 year PPA, from which grants of US$50,000 each are disbursed each year to benefit environmental and social activities in Kenya's Coast Region. Iberafrica has a social responsibility programme, and IPTL also is an active donor to its immediate community. Jorf Lasfar received the American Chamber of Commerce award for community development and developed an ash disposal facility (previously ash was pumped into the sea). CMS's social responsibility involvement in Ghana included providing scholarships for secondary and tertiary education as well as providing support for both medical clinics and the construction of drainage systems. Although the sums are not significant, these programmes, particularly when well advertised, have the potential to win allies and counter the stereotype of IPPs.

Another perhaps more significant action is how sponsors cope with stress, such as macroeconomic shock and associated currency devaluation or pressure from host governments who perceive costs to be too high. Although anecdotal, there is evidence for how such strategic management aided the Egyptian plants, which may be judged among the most successful in this pool of IPPs. There is also evidence for how strategic management helped put Kenya's Iberafrica back on track, in contrast to Westmont, where no such management is in evidence. Finally, both of Tanzania's IPPs may accredit future sustainability in part to such an element.

In the case of the Egyptian devaluation, EEHC appears to have approached Sidi Krir management when the country was experiencing an acute scarcity of dollars to request payment in pounds to the maximum extent feasible, but that due to its US dollar denominated debt the firm was unable to acquiesce. Minor changes, since the devaluation, are limited to partial payment of the local operating and maintenance component (both fixed and variable) in local currency, which amounts to approximately 4 per cent of the total charge. With Sidi Krir, the change is an informal agreement between the project's general manager and EEHC. With Port Said and Suez, the agreement has gone through negotiations with IFC, but EDF could have chosen to return to US dollar payments at any time, i.e. it was not contractually bound. Although small, these actions do send the message that sponsors are willing to work where possible to make the situation less onerous for the host country.

Kenya's Iberafrica has dealt with two stresses: drought and alleged corruption. It is important to reiterate in this context that the project was also up for a contract renewal at the time when the following actions described were undertaken. According to stakeholders at Iberafrica, the IPP voluntarily lowered its capacity charge at a time when KPLC was operating in the red, due in part to a drought-related recession, to show its support for the country and signal its interest in a second contract (the first lasted seven years). Iberafrica later secured a second contract, albeit after even further reductions were negotiated and passed by the electricity regulator.[122]

A final area which may yield greater balancing in terms of development and investment outcomes is in the refinancing of projects, evidence for which one may see as Songas seeks to refinance its 100 per cent equity investment in the second phase of the plant. Possible refinancing in the case of IPTL with the Government of Tanzania proposing to buy the outstanding debt, and possibly equity, could also lead to what may be perceived as more balanced outcomes. If and when the Government of Tanzania buys back IPTL, it will make a once-off payment on behalf of the utility and then pass the asset ownership to TANESCO, which may subsequently decide to convert the plant to run on natural gas. Through this transaction, the capacity charge will be reduced to a token amount, and following gas conversion, the energy charge dropped from US$9-12 million to US$1-1.5 million per month. The PPA will be terminated, a new agreement drafted, and the customers will see discounted tariffs.

Refinancing does, however, only have limited application, and must be dealt with carefully during the project negotiation for as one banker candidly indicated: “if project finance bankers are expected to finance projects with the understanding that periodically it will be necessary to have a restructuring, the outcome of which is uncertain, the result will be to eliminate the availability of non-recourse financing”—which, given the already low levels in Africa should be avoided.[123]

It is the government’s willingness to share risks over the life of the project, which may also be pivotal in the long-term sustainability of projects. Strategic management does not occur in a vacuum, with the sponsor alone. Often the host country government may not only be an active counterparty, but even, as evidenced in the refinancing of IPTL, initiate such strategic management. Other government led initiatives include, among others, Morocco’s ONE assuming a greater equity share in Tahaddart when EDF pulled out and the Government of Tanzania’s buying down of Songas’ AFUDC.

The myriad project level factors are summarized in the table immediately below.

[122] It should, however, be noted that the fact that Iberafrica was not project-financed meant that the company was at greater liberty to change the payment streams.

[123] See footnote 108 for indication of low project finance levels in Africa.

Table 4: Contributing elements to successful IPP investments, project issues

CES	Details
Favourable equity partners	-Local capital/partner contribution, where possible -Risk appetite for project -Experience with developing country project risk -Involvement of a DFI partner (and/or host country government) -Reasonable, fair ROE -Development-minded firms
Favourable debt arrangements	-Low cost financing -Local capital/markets mitigate foreign exchange risk -Risk premium demanded by financiers or capped by off-taker matches country/project risk -Some flexibility in terms and conditions (possible refinancing)
Secure and adequate revenue stream	-Commercially sound metering, billing and collections by the utility - Robust PPA (stipulates capacity and payment as well as dispatch, fuel metering, interconnection, insurance, *force majeure*, transfer, termination, change of law provisions, refinancing arrangements, dispute resolution, etc.) - Security arrangements where necessary (escrow accounts, letters of credit, stand-by debt facilities, hedging and other derivative instruments, committed public budget and/or taxes/levies, targeted subsidies and output-based aid, hard currency contracts, indexation in contracts)
Credit enhancements and other risk management and mitigation measures	-Sovereign guarantees -Political risk insurance -Partial risk guarantees -International arbitration
Positive technical performance	-Technical performance high (including availability) -Sponsors anticipate potential conflicts (especially related to O&M, and budgeting) and mitigate them
Strategic management and relationship building	Sponsors work to create good image in country through political relationships, development funds, effective communications and strategically manage their contracts, particularly in the face of exogenous shocks and other stresses

Notes: ROE: return on equity; O&M: operation and maintenance

5. Conclusion

In concluding, on the one hand, where there was a perceived balance between sponsors and host country governments, contracts generally remained intact, as seen in most of the North African cases, with the contributing elements of success being: the more favourable country level factors (such as favourable investment climates, clear policy frameworks, and ICBs, among others). On the other hand, perceived imbalances (often exaggerated by exogenous stresses) between sponsors and host country governments frequently did lead to an unravelling of the original contract. Neither the PPA, nor the security arrangements were sufficient in locking in long-term sustainability.

Although the evidence is not conclusive, strategic management on behalf of sponsors and government as well as strong technical performance have been used to cope with contract instability. Furthermore, the fact that projects with participation of development-minded firms and DFIs were less likely to unravel signals two points: such projects may have been more balanced from the get-go, and when an exogenous stress struck, they may have also been better equipped to resist any form of host country government pressure.

Thus, the findings are four-fold. First, evidence for contract unravelling is widespread across the pool of African IPPs where an imbalance is perceived, which largely corresponds to the more risky SSA projects. Secondly, the incidence of such unravelling does not necessarily signal the end of a project's operation. New agreements may be reached, albeit at a cost, that prove sustainable. Third, efforts must continue to close the initial gap between investors and host country governments (or else examples of the further contract unravelling will continue). Finally, the means to closing the gap may not be only, or mainly, be via increasing the sort of new protections, including political risk insurance, which have been reported to often confound political and economic issues and may instead lie in systematic treatment of the numerous contributing elements to success defined by this paper.

Appendix A

Detailed specifications are presented for each IPP discussed in this thesis. The organization of the data is as follows. Information is first grouped into regions with North Africa, followed by East Africa and West Africa. Countries within each of the regions are presented in alphabetical order. Under each country, IPPs are then presented in terms of chronological order, based on the date when the IPP first came online, as per the outline below.

Region/Country/Project	**Page numbers**
North Africa	75-81
Egypt	75-76
Sidi Krir	75
Port Said & Suez	76
Morocco	77-79
Jorf Lasfar	77
Compagnie Eolienne de Detroit (CED)	78
Tahaddart	79
Tunisia	80-81
Rades II	80
Societé d'Electricité d'El Biban (SEEB)	81
East Africa	82-88
Kenya	82-85
Westmont	82
Iberafrica	83
OrPower4	84
Tsavo	85
Tanzania	86-88
IPTL	86
Songas	87
Mtwara	88
West Africa	89-93
Cote d'Ivoire	89-90
Compagnie Ivoirienne De Production D'Electricite (CIPREL)	89
Azito	90
Ghana	91
Takoradi II	91
Nigeria	92-93
AES Barge Limited	92
Okpai	93

North Africa

Egypt (1 of 3 projects)

Project	**Sidi Krir**
Size	683 MW
Cost	US$413.9 million[124]
$ per kW	606
Technology	Natural gas/steam cycle
ICB	Yes
Contract	BOOT, 20 years
Debt/Equity	33/67
DFI in equity and debt	None
Local participation in equity and debt	Yes (equity)
Equity partners (country headquarters & % of each)	InterGen (USA, 60%) & Edison (USA, 40%) sold to Globeleq (UK, 100%) in 2004-2005, conditional sale to Pendekar Energy Limited between Malaysia Tanjong & Saudi Arabia AlJomaih, 100%) in 2007
Lenders	60% sourced from local Egyptian banks, led by Commercial International Bank, with remaining 40% provided by international commercial banks
Credit enhancements	Sovereign guarantee
Security arrangements	None
Project tender, COD	1996, 2002
Contract change	None
Fuel arrangement	Fuel provided by Gasco, national gas company, at sub-economic rates, project sponsors pay Gasco and are subsequently reimbursed by EEHC, i.e. not a direct pass through

[124] Previously project costs were estimated at US$417.8, however, following negotiations with Egyptian customs authorities costs were reduced to US$413.9.

Egypt (2 & 3 of 3 projects[125])

Project	**Port Said & Suez**
Size	683 MW each (for a total of 1366)
Cost	US$340 million (Port Said) and US$338 million (Suez)
$ per Kw	498, 495
Technology	natural gas/steam cycle
ICB	Yes
Contract	BOOT, 20 years
Debt/Equity	25/75
DFI in equity and debt	Yes (debt)
Local participation in equity and debt	None
Equity partners (country of origin & % of each shareholder)	EDF (France, 100%) sold to Kuasa Power, subsidiary of Tanjong (Malaysia, 100%) in 2006
Lenders	IFC, Societe Generale, Barclays, and John Hancock (the USA-based insurance firm)
Credit enhancements	Sovereign guarantee
Security arrangements	None
Project tender, COD	1998, 2002 (Port Said) and 1998, 2003 (Suez)
Contract change	None
Fuel arrangement	Fuel provided by Gasco, national gas company, at sub-economic rates, project sponsors pay Gasco and are subsequently reimbursed by EEHC, i.e. not a direct pass through

[125] Port Said and Suez are grouped under one profile as the project size, type and sponsor were identical. The only distinction between the two plants was in the capital cost, with Suez, which was negotiated after Port Said, recording a lower capital cost.

Morocco (1 of 3 projects)

Project	**Jorf Lasfar**
Size	680+680[126] MW
Cost	US$1500 million[127]
$ per kW	US$1,103
Technology	Coal
ICB	Yes
Contract	BTO, 30 years
Debt/Equity	67/33
DFI in equity and debt	Yes (debt)
Local participation in equity and debt	None
Equity partners (country of origin & % of each shareholder)	CMS (USA, 50%) and ABB (Swiss, 50%) sold shares to TAQA (UAE, 100%) in 2007
Lenders	USA ExIm (US$200 million), OPIC and a syndicate of approximately 50 commercial banks
Credit enhancements	PRG from the Italian Export Credit Agency (SACE), the Swiss Export Risk Guarantee (ERG) and the World Bank
Security arrangements	Escrow account equivalent to 1 month capacity charge
Project tender, COD	1994, 2000
Contract change	None
Fuel arrangement	Coal procured by project, with 80% of coal costs reimbursed at the average cost of coal procured by project and 20% of costs reimbursed at average cost of coal imported into European Union

[126] The first 680 MW refer to an existing facility, i.e. a brownfield investment; the greenfield investment was an additional 680 MW.
[127] The project costs included the upgrade of the coal receiving facility and also the infrastructure for transporting coal to ONE's Mohamedia plant.

Morocco (2 of 3 projects)

Project	**Compagnie Eolienne de Detroit (CED)**
Size	50 MW
Cost	$58.5 million[128]
$ per kW	$1,170
Technology	Wind
ICB	Yes
Contract	BTO, 19 years
Debt/Equity	70/30
DFI in equity and debt	Yes (debt)
Local participation in equity and debt	None
Equity partners (country of origin & % of each shareholder)	EDF (France, 49%), Paribas (France, 35.5%) & GERMA (France, 15.5%) maintained equity since 1997
Lenders	EIB (€24.4 million), PROPARCO, Credit Agricole (now Calyon) and other commercial banks
Credit enhancements	None
Security arrangements	None
Project tender, COD	1995, 2000
Contract change	None
Fuel arrangement	No energy charge, wind powered

[128] Project costs quoted in Euros at €45.7, converted to US$ on August 24 2006 at exchange rate of €1=US$1.28.

Morocco (3 of 3 projects)

Project	**Tahaddart**
Size	384 MW
Cost	US$364.9 million
$ per kW	US$950
Technology	Natural gas/combined cycle
ICB	Yes
Contract	BTO, 20 years
Debt/Equity	75/25
DFI in equity and debt	None
Local participation in equity and debt	Yes (equity and debt)
Equity partners (country of origin & % of each shareholder)	ONE (Morocco, 48%), Endesa (Spain, 32%) and Siemens (Germany, 20%) maintained equity since 1999
Lenders	Banque Centrale Populaire, Banque Marocaine pour le Commerce Exterieure, Credit Agricole
Credit enhancements	None
Security arrangements	Letter of Credit equivalent to 1 month capacity charge
Project tender, COD	1999, 2005
Contract change	None
Fuel arrangement	Government supplies gas (which is received as a royalty from the Algerian Spain pipeline)

***Tunisia** (1 of 2 projects)*

Project	Rades II
Size	471 MW
Cost	US$260.7 million
$ per kW	US$554
Technology	Natural gas/combined cycle
ICB	Yes
Contract	BOO, 20 years
Debt/Equity	70/30
DFI in equity and debt	Yes (debt)
Local participation in equity and debt	None
Equity partners (country of origin & % of each shareholder)	PSEG (USA, 60%) sold shares to BTU Power Company (GCC, 60%) in 2004, Marubeni (Japan, 40%) retained shares since 1999
Lenders	JBIC, BNP Sanwa and other commercial banks
Credit enhancements	No sovereign guarantee but did receive reassurances from government that STEG would be able to meet its obligations; JBIC also received Letter of Comfort from Ministry of Energy indicating government committed to success of project
Security arrangements	Escrow account (amount not publicly available)
Project tender, COD	1997, 2002
Contract change	None
Fuel arrangement	Fuel supplied by STEG and fuel costs are a pass through to STEG

Tunisia (2 of 2 projects)

Project	Societe d'Electricite d'El Biban (SEEB)
Size	27 MW
Cost	US$30 million[129]
$ per kW	US$1,111
Technology	Natural gas/open cycle
ICB	None, but selective international tender conducted
Contract	BOO, 20 years
Debt/Equity	67/33
DFI in equity and debt	None
Local participation in equity and debt	Yes (debt, but US$ denominated)
Equity partners (country of origin & % of each shareholder)	Centurion (USA, 50%) sold to Candex (Canada, 50%) in 2005, Caterpillar (USA, 50%) maintained equity since 2002
Lenders	ABC International Bank (50%), AMEN Bank (50%)
Credit enhancements	None
Security arrangements	None
Project tender, COD	2000, 2003
Contract change	Project has not been able to secure its fuel supply and is presently offline
Fuel arrangement	Project responsible for providing fuel

[129] Project costs include the gas infrastructure.

East Africa

***Kenya** (1 of 4 projects)*

Project	**Westmont**
Size	46 MW
Cost	US$65 million
$ per kW	US$435
Technology	Kerosene/gas condensate/gas turbine (barge mounted)
ICB	None, but selective international tender conducted
Contract	BOO, 7 years
Debt/Equity	NA
DFI in equity and debt	None
Local participation in equity and debt	None
Equity partners (country of origin & % of each shareholder)	Westmont (Malaysia, 100%) has sought to sell plant since 2004
Lenders	NA
Credit enhancements	None
Security arrangements	None
Project tender, COD	1996, 1997
Contract change	Not a contract change per se, but firm failed to negotiate a second contract after its 7 year contract ended in 2004 due to failure to agree on tariffs
Fuel arrangement	Originally Westmont to procure fuel and then pass through to utility, however, following dispute with fuel supplier about taxes after the first year of operation, utility took over procurement

Kenya (2 of 4 projects)

Project	**Iberafrica**
Size	56 MW
Cost	US$35 million
$ per kW	US$1,161
Technology	HFO/medium speed diesel engine
ICB	None
Contract	BOO, 7 years + 15 years[130]
Debt/Equity	72/28
DFI in equity and debt	None
Local participation in equity and debt	Yes (equity and debt)
Equity partners (country of origin & % of each shareholder)	Union Fenosa (Spain, 80%), KPLC Pension Fund (Kenya, 20%) since 1997
Lenders	Union Fenosa (US$12.7 million in direct loans and guaranteed US$20 million); KPLC Staff Pension Fund (US$9.4 in direct loans and guaranteed US$5 million through a local Kenyan bank).
Credit enhancements	None
Security arrangements	None
Project tender, COD	1996, 1997
Contract change	Yes, Iberafrica reduced the capacity charge of its first PPA by 37 per cent in April 2002 and then to 59 per cent of the original PPA in September 2003.[131]
Fuel arrangement	Iberafrica buys fuel and passes cost through to KPLC based on the units generated and specific consumption parameters agreed on in the PPA

[130] Iberafrica has had two different PPAs, the first spanning 7 years, the second, signed in 2004, which will span 15 years.

[131] Furthermore, although not a contract change per se, the value of the capacity charge for Iberafrica's second PPA was 50 per cent that of the first PPA.

Kenya (3 of 4 projects)

Project	OrPower4
Size	13 MW
Cost	US$54 million
$ per kW	US$4,154
Technology	Geothermal
ICB	Yes
Contract	BOO, 20 years
Debt/Equity	NA
DFI in equity and debt	None
Local participation in equity and debt	None
Equity partners (country of origin & % of each shareholder)	Ormat (100%) since 1998
Lenders	NA
Credit enhancements	MIGA guarantee
Security arrangements	Escrow account, one month capacity charges, and a stand-by Letter of Credit, covering three months billing (although only finalized at end-2006)
Project tender, COD	1996, 2000
Contract change	Yes, tariff for the second phased (35 MW) reduced
Fuel arrangement	The only fuel arrangement per se is that OrPower4 leases the wells from the government, to which it pays a royalty of sorts

Kenya (4 of 4 projects)

Project	**Tsavo**
Size	75 MW
Cost	US$85 million
$ per kW	US$1,133
Technology	HFO/medium speed diesel engine
ICB	Yes
Contract	BOO, 20 years
Debt/Equity	78/22
DFI in equity and debt	Yes (equity and debt)
Local participation in equity and debt	None
Equity partners (country of origin & % of each shareholder)	Cinergy & IPS jointly owned 49. 9%, Cinergy sold to Duke Energy in 2005, CDC/Globeleq (UK, 30%), Wartsila (Finland, 15%), IFC (5%) retain remaining shares since 2000
Lenders	IFC own account (US$16.5 million), IFC syndicated (US$23.5 million); CDC own account (US$13 million); DEG own account (€11 million), DEG syndicated (€2 million)
Credit enhancements	Letter of Comfort provided by government
Security arrangements	Escrow account, equivalent to 1 month capacity charge, and a stand-by Letter of Credit, equivalent to 3 months billing
Project tender, COD	1995, 2001
Contract change	None, but was pressured to lower tariff
Fuel arrangement	Tsavo buys fuel and passes cost through to KPLC based on the units generated and specific consumption parameters agreed on in the PPA

Tanzania (1 of 3 projects)

Project	IPTL
Size	100 MW
Cost	US$120 million
$ per kW	US$1,200
Technology	HFO/medium speed diesel engine
ICB	None
Contract	BOO, 20 years
Debt/Equity	70/30
DFI in equity and debt	None
Local participation in equity and debt	Yes (equity)
Equity partners (country of origin & % of each shareholder)	Mechmar (Malaysia, 70%), VIP (Tanzania, 30% **in kind**), both have sought to sell shares
Lenders	Two Malaysian-based banks (Bank Bumiputra Malaysia Berhad--now Bank Bumiputra Commercial Bank--and SIME Bank)
Credit enhancements	Sovereign guarantee
Security arrangements	Liquidity facility equivalent to 4 months capacity charge (but not yet established)
Project tender, COD	1997, 1998
Contract change	Yes, post arbitration, monthly capacity charges lowered from US$3.6 million to US$2.6 million
Fuel arrangement	IPTL imports fuel, which is a pass through to the utility

Tanzania (2 of 3 projects)

Project	**Songas**
Size	180 MW[132]
Cost	US$316 million[133]
$ per kW	US$2,313 (for first 115 MW) and US$769 (for 65 MW expansion)
Technology	Natural gas/open cycle
ICB	Yes
Contract	BOO, 20 years
Debt/Equity	70/30 for 115 MW 100% equity financed for 65MW expansion
DFI in equity and debt	Yes
Local participation in equity and debt	Yes (TANESCO, TDFL)
Equity partners (country of origin & % of each shareholder)	TransCanada sold majority shares to AES (USA) in 1999 and AES sold majority shares to Globeleq (UK) in 2003[134]
Lenders	World Bank (US$136 million), EIB, (US$55 million), Sida, (US$15 million)
Credit enhancements	None
Security arrangements	Escrow account: for first 115 MW, with the government matching every US$1 spent by the project company; liquidity facility equivalent to 4 months capacity charge for the first 3 years, declining to 2 months starting in year 4 through the remaining years of the contract
Project tender, COD	1994, 2004
Contract change	Yes, the buying down of Songas' AFUDC (of approximately US$103 million) by the Government of Tanzania cut the monthly capacity charge by about one third.[135]
Fuel arrangement	Songo Songo gas provided to project company at a rate of US$0.55/MMBtu for turbines I-V and at US$2.17 MMBtu for turbine VI

[132]There was considerable evolution in terms of the planned capacity for the plant, from 60 MW to the current 180 MW.

[133]Songas project costs include refurbishment of gas wells, a new gas processing facility, pipeline construction and fuel conversion of the existing power station (Ubungo), in total amounting to US$266 million, and an additional US$50 million for expansion in terms of two additional turbines (total 65 MW) and related infrastructure. The expansion was financed entirely by equity. A rough estimate for the electricity generation component would be 40 per cent of project costs or US$130 million, based on US$35 million for refurbishment and fuel conversion of existing turbines, US$45 million assumed loans on existing turbines, and US$50 million for expansion.

[134] Due to complexity, turnover is detailed in this footnote: Ocelot (Canada), TransCanada (Canada), Tanzania Petroleum Development Corporation, TPDC (Tanzania), TANESCO (Tanzania), Tanzania Development Finance Company, TDFL (Tanzania, sponsored by EIB), IFC (multilateral), DEG (German), CDC (UK) were shareholders by 1996, with TransCanada the majority shareholder; IFC and DEG sold shares to CDC in 1997/8; TransCanada sold shares to AES (USA) in 1999; Ocelot/PanOcean sold shares to AES in 2001; AES sold majority shares to Globeleq (UK) and FMO (Holland) in 2003. *After the AES sale, equity shares and associated financial commitments (expressed in US$ million) in Songas were as follows: Globeleq: US$33.8 (56%); FMO: US$14.6 (24%); TDFL: US$4 (7%); CDC: US$3.6 (6%); TPDC: US$3 (5%) and TANESCO: US$1 (2%). This does not reflect the additional US$50 million that Globeleq committed for the expansion.*

[135]The buying down of the AFUDC, which was negotiated by the Government of Tanzania during AES's sale to Globeleq, has not, however, negatively impacted on the sponsor since the full amount of the buy-down was paid to Globeleq. In addition, since May 2005, TANESCO has not been paying the portion of the capacity charge that relates to the project's subordinated debt. Although such an arrangement was provided for in the subsidiary loan agreement of 2001, what this means is that the IDA credit that the Government of Tanzania on-lent to TANESCO at a higher rate is presently not being serviced by the utility.

Tanzania (3 of 3)

Project	**Mtwara Energy Project**
Size	12 MW
Cost	US$ 8.2 million
$ per kW	US$683
Technology	gas-fired reciprocating engine
ICB	None
Contract	BOO, 2 years (20 years expected)
Debt/Equity	100% financed with balance sheet of shareholders to date
DFI in equity and debt	Yes (equity)
Local participation in equity and debt	None
Equity partners (country of origin & % of each shareholder)	Artumas (Canada, 80%), FMO (Netherlands, 20%)
Lenders	100% financed with balance sheet of shareholders to date
Credit enhancements	None
Security arrangements	Escrow account during interim PPA (2 years), during 20 year PPA, Tariff Equalization Fund expected, a fixed-value account designed to make up the difference between the national tariff and the cost of power under the project
Project tender, COD	1994, 2007
Contract change	None
Fuel arrangement	Fuel provided by sponsor, at a charge of US$5.00 per MMBtu, which is passed through to utility

West Africa

Cote d'Ivoire (1 of 2)

Project	**Compagnie Ivoirienne de Production d'Electricité (CIPREL)**
Size	210 MW
Cost	US$105.6 million[136]
$ per kW	US$503
Technology	Natural gas/open cycle
ICB	None
Contract	BOOT, 19 years
Debt/Equity	-
DFI in equity and debt	Yes (equity and debt)
Local participation in equity and debt	None
Equity partners (country of origin & % of each shareholder)	SAUR International, with 88% (JV between French SAUR Group owned by Bouygues, 65% and EDF, 35%) BOAD, PROPARCO, and IFC holding the remaining 12%; in 2005 all shares sold to Bouygues (France, 98%), except BOAD (2%)
Lenders	World Bank
Credit enhancements	None
Security arrangements	None
Project tender, COD	1993, 1995
Contract change	None
Fuel arrangement	Government procures fuel

[136] Investment cost €87.8m or 57.6 billion CFA, with the average 1994 conversion of US$ to CFA, 545.100.

Cote d'Ivoire (2 of 2)

Project	**Azito**
Size	330 MW[137]
Cost	US$233 million
$ per kW	US$706
Technology	Natural gas/open cycle
ICB	Yes
Contract	BOOT, 24 years
Debt/Equity	70/30
DFI in equity and debt	Yes (debt)
Local participation in equity and debt	None
Equity partners (country of origin & % of each shareholder)	Cinergy (JV between Swiss ABB, 50% and French EDF, 50%) Holds 65.7% of shares, CDC/Globeleq (11%), and IPS (23%)
Lenders	IFC, CDC, FMO, DEG, AfDB, Societe Generale and other European commercial banks
Credit enhancements	World Bank partial risk guarantee Sovereign guarantee
Security arrangements	Escrow account equivalent to 1 month capacity charge
Project tender, COD	1996, 2000
Contract change	None
Fuel arrangement	Government procures fuel

[137] The initial project concept included specifications to raise capacity to 420 MW.

Ghana (1 of 1)

Project	**Takoradi II**
Size	220 MW[138]
Cost	US$ 110 million
$ per kW	US$500
Technology	Light crude oil/single cycle
ICB	None
Contract	BOOT, 25 years
Debt/Equity	Financed exclusively with balance sheet of sponsors
DFI in equity and debt	No
Local participation in equity and debt	Yes (equity)
Equity partners (country of origin & % of each shareholder)	CMS (USA, 90%), VRA (Ghana, 10%), CMS sold shares to TAQA (UAE, 90%) in 2007
Lenders	Financed exclusively with balance sheet of sponsors
Credit enhancements	Sovereign guarantee
Security arrangements	US$ 3 million Letter of Credit provided by government
Project tender, COD	1998, 2000
Contract change	Failure to develop 2nd phase (110 MW), investors cite the lack of government guarantees (granted in the first phase of Takoradi II) meanwhile government has indicated that the EPC costs are too high and further development remains at standstill
Fuel arrangement	Government procures fuel

[138] The initial project concept included specifications to add a second phase of 110 MW and convert to combined cycle, however, lack of funding has limited the completion of this phase.

Nigeria (1 of 2)

Project	**AES Barge Limited**
Size	270 MW
Cost	US$240
$ per kW	
Technology	Natural gas/open cycle (barge mounted)
ICB	None
Contract	BOO, 20 years
Debt/Equity	NA
DFI in equity and debt	None
Local participation in equity and debt	Yes (equity)
Equity partners (country of origin & % of each shareholder)	Enron (USA, 100%) sold to AES (95%) and YFP (Nigeria, 5%) in 2000
Lenders	NA
Credit enhancements	OPIC political risk insurance Sovereign guarantee
Security arrangements	US$60 million Letter of Credit from Ministry of Finance
Project tender, COD	1999, 2001
Contract change	Yes, initial plant size increased from 90 MW to 270 MW (9 units of 30 MW each) and change in the fuel from liquid fuel to natural gas, both of which had the effect of reducing the capacity charge, current contract changes under discussion involve among other things the availability deficiency payment, meanwhile tax exemption certificate has been withheld by government for the duration of the project
Fuel arrangement	Utility arranges fuel

Nigeria (2 of 2)

Project	**Okpai**
Size	450 MW
Cost	US$ 462[139]
$ per kW	
Technology	Natural gas/combined cycle
ICB	None
Contract	BOO, 20 years
Debt/Equity	100% equity financed
DFI in equity and debt	None
Local participation in equity and debt	Yes (equity and debt)
Equity partners (country of origin & % of each shareholder)	Nigerian National Petroleum Corporation (Nigeria, 60%), Nigerian Agip Oil Company (Italy, 20%), and Phillips Oil Company (USA, 20%) maintained equity since 2001
Lenders	Provided by equity partners
Credit enhancements	PPA backed by Nigerian Petroleum Development Company's oil revenues
Security arrangements	None
Project tender, COD	2001, 2005
Contract change	Ongoing negotiations related to investment costs which rose by US$150 million, to US$462 million; although plant is producing power, due to the dispute, full payment is not being made by utility
Fuel arrangement	Project company provides fuel

[139] Project costs include the gas infrastructure.

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